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The Emergence and Proliferation of Free Trade Agreements in East Asia

Globalization of economic activities has been accelerating rapidly as cross-border movements of goods, money, information, and people have expanded remarkably in recent years. Although globalization has allegedly brought some negative consequences such as the widening of income gaps among countries as well as within countries, on the whole it has brought positive impacts such as rapid economic growth and technological progress. One factor that has contributed to globalization is substantial trade liberalization, carried out under the auspices of the General Agreement on Tariffs and Trade/World Trade Organization (GATT/WTO).

Against the background of rapid globalization, regionalization (or regional economic integration) has emerged in various parts of the world in recent years. Regionalization takes two forms. One type of regionalization arises as a result of natural economic developments in that the benefits of agglomeration including economies of scope, scale, and speed outweigh the costs of agglomeration, such as congestion. Indeed, rapidly growing economies in proximity interact with each other through market and nonmarket channels to accelerate their economic growth. The other type of regionalization involves institutional arrangement such as regional trade agreements (RTAs), including free trade agreements (FTAs) and customs union. RTAs are discriminatory trade agreements, providing only members with preferential treatment. The former type of regionalization may be characterized as "market-driven" and the latter as "institution-driven."

Rapid economic growth has increased East Asia's importance in the world economy in the post–World War II period. As a consequence, East Asia, along with North America and Western Europe, has come to be recognized as an important region that influences the world economy. Regional economic integration in Western Europe started in the early post–World War II period, mainly in the form of establishing regional institutions, while regional economic integration in North

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America was initially developed through market forces, and then was supplemented by a regional institution. As such, regionalization in Western Europe may be characterized as "institution-driven," while regionalization in North America was at first "market-driven" in the early stages then supplemented by "institution-driven" regionalization. In light of these global and regional developments, this article examines patterns of regional economic integration in East Asia. From the analysis we observe a shift from "market-driven" to "institution-driven" regionalization in East Asia, similar to North America.

Rapid Expansion of Foreign Trade and Foreign Direct Investment in East Asia

Foreign trade and foreign direct investment have played important roles in international economic activities worldwide and in particular in East Asia.[1] Indeed, rapid expansion of foreign trade and FDI has been considered a crucial factor contributing to rapid economic growth in East Asia. This section reviews globalization of economic activities in East Asia by focusing on the changing patterns of foreign trade and FDI in the 1980s and 1990s, in order to set the stage for the analysis of the changing nature of regionalization from market-driven to institution-driven types toward the end of the 1990s.

Foreign Trade

The foreign trade of East Asian economies expanded rapidly beginning in the mid-1980s until the outbreak of the East Asian financial crisis (Table 1 and Figure 1). Between 1985 and 1997, exports from emerging East Asian economies expanded steadily to register an almost fivefold increase, before declining in 1998 as a result of the crisis. The rate of expansion was particularly high from 1986 to 1988, when the annual rate of growth exceeded 20 percent. The 1990s saw fluctuations in the annual rates of growth, with a peak at 22 percent in 1995 followed by a decline, resulting in negative growth in 1998. The patterns of export growth for the 1985–99 period are similar for all emerging East Asian economies, with few exceptions. Compared with the four newly industrialized economies (NIE-4, i.e., Hong Kong, Korea, Singapore and Taiwan) and the core Association of Southeast Asian Nations (ASEAN) economies (Indonesia, Malaysia, the Philippines, and Thailand), China registered a significantly higher growth rate in the 1990s. Compared with the emerging East Asian economies, Japan fared less favorably in terms of export expansion, as Japan's exports increased only 2.4-fold between 1985 and 1997.

As a result of rapid export expansion, East Asia increased its share of world exports from 9 percent in 1980–85 to 18 percent in 1997, before declining slightly in 1998. As a group, the NIE-4 expanded their share from 5 percent in 1980–85 to 10 percent in 1997. China became the largest exporter among the emerging East

Asian economies in 1999, accounting for 3 percent of world exports. Although still maintaining a substantial share in world exports, Japan's share in world exports declined from 10.3 percent in 1986 to 7.2 percent in 1998 because of relatively slower growth.

One notable development was the rapid expansion of manufactured exports. Specifically, the share of manufactured products in total exports for the NIE-4 and the ASEAN-5 increased from 71 percent and 18 percent, respectively, in 1980 to 87 percent and 60 percent in 1995. For China, the corresponding share increased from 67 percent in 1990 to 79 percent in 1995.

Like exports, imports to emerging East Asian economies increased substantially in the 1980s and 1990s before the crisis. Specifically, their imports increased 6.5 times between 1980 and 1997, before a decline in 1998 due to the crisis. Among the emerging East Asian economies, Hong Kong achieved an especially large increase of almost tenfold during the 1980–97 period. Hong Kong is followed by China and Malaysia, whose imports increased 7.3-fold. In contrast to the emerging East Asian economies, which achieved remarkable import expansion, Japan saw its imports only double in seventeen years from 1980 to 1997. Indeed, it should be noted that the magnitude of Japan's imports was comparable to that of the emerging East Asian economies in the early 1980s, but in the late 1990s it was only a little less than one-third of the imports of the emerging East Asian economies.

The rapid expansion of imports by the emerging East Asian economies resulted in an increase in their share of world imports from 8.1 percent in 1980 to 18 percent in 1997. By contrast, Japan's share in world imports declined from 7.4 percent in 1980 to 5.1 percent in 1998, before a slight increase in 1999.

In summary, an examination of foreign trade by East Asian economies revealed a substantial increase in both exports and imports in the 1980s and 1990s before they were struck by the financial crisis in 1997. We also observed somewhat contrasting developments between the emerging East Asian economies, on the one hand, and Japan, on the other hand. Emerging East Asian economies, particularly China in the 1990s, registered remarkable growth, while Japan achieved a less spectacular performance. As a result of rapid trade expansion by East Asian economies, their share in world exports and imports increased from 15 percent and 15 percent, respectively, in 1980 to 25 percent and 21 percent in 1999, respectively. A large part of the increase was achieved by the emerging East Asian economies, although Japan still has a substantial position, amounting to about one-third of trade conducted by all East Asian economies. It is worth noting that the magnitude of exports by East Asian economies was twice as large as the magnitude for the United States and approximately 65 percent of the value for the European Union (EU) in 1999, while the corresponding values in terms of imports were 114 percent and 56 percent, respectively. These observations indicate that East Asia represents a significant position in world trade.

Table 1

International Trade of East Asian Economies in the 1980s and 1990s

(a) Value in millions of U.S. dollars

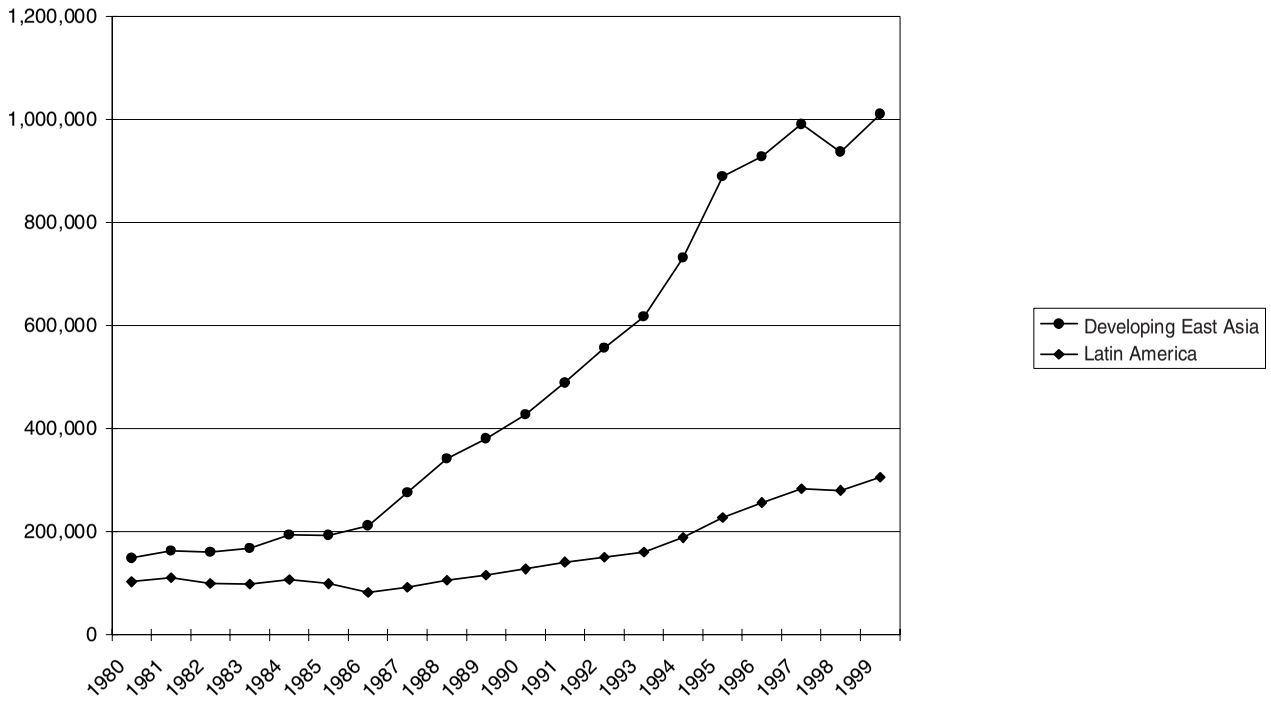
Regions/ countries	Exports					Imports					Total trade (exports + imports)				
	1980	1985	1990	1995	1999	1980	1985	1990	1995	1999	1980	1985	1990	1995	1999
World	1,832,508	1,874,505	3,381,690	5,070,827	5,663,310	1,918,724	1,975,924	3,517,259	5,137,927	5,810,326	3,751,232	3,850,429	6,898,949	10,208,754	11,473,636
United States	220,781	213,146	393,106	583,451	690,689	256,959	361,620	517,020	770,972	1,048,435	477,740	574,767	910,126	1,354,423	1,739,124
European Union-15	751,159	708,197	1,488,365	2,018,290	2,208,491	843,944	724,609	1,538,964	1,914,173	2,143,856	1,595,103	1,432,805	3,027,329	3,932,463	4,352,347
East Asia	279,576	370,825	715,547	1,332,658	1,429,911	292,342	323,547	669,517	1,267,960	1,202,865	571,918	694,372	1,385,064	2,600,618	2,632,776
Japan	130,435	177,189	287,678	443,047	419,207	141,284	130,516	235,307	336,027	310,733	271,719	307,705	522,985	779,074	729,940
Emerging East Asia	149,141	193,636	427,869	889,611	1,010,704	151,058	193,031	434,210	931,932	892,132	300,199	386,667	862,079	1,821,544	1,902,836
NIE-4	76,262	113,966	269,222	529,143	553,671	88,148	107,119	271,594	556,160	521,150	164,410	221,085	540,815	1,085,303	1,074,821
Hong Kong SAR	19,720	30,182	82,144	173,556	173,793	22,399	29,701	82,482	192,764	179,650	42,119	59,883	164,626	366,321	353,443
Korea	17,439	30,289	67,812	125,588	143,647	22,063	31,058	74,405	135,352	119,740	39,502	61,347	142,217	260,940	263,387
Singapore	19,377	22,812	52,753	118,187	114,730	24,013	26,237	60,954	124,394	111,071	43,390	49,049	113,707	242,581	225,801
Taiwan	19,726	30,683	66,513	111,813	121,501	19,673	20,123	53,753	103,649	110,689	39,399	50,806	120,266	215,462	232,191
ASEAN-9	52,162	49,692	91,612	204,417	255,046	40,525	39,999	102,289	235,746	197,636	92,686	89,691	193,902	440,163	452,682
Indonesia	21,909	18,596	25,675	45,428	57,193	10,837	10,275	22,005	40,629	28,950	32,747	28,871	47,680	86,056	86,142
Malaysia	12,960	15,408	29,420	73,724	84,550	10,821	12,301	29,170	77,620	65,491	23,782	27,709	58,590	151,344	150,042
Philippines	5,787	4,614	8,194	17,371	35,474	8,295	5,351	12,993	28,282	31,368	14,082	9,965	21,186	45,653	66,842
Thailand	6,501	7,123	23,072	57,201	61,797	9,213	9,259	33,407	73,692	53,207	15,714	16,382	56,479	130,892	115,003
Vietnam	0	693	2,525	5,450	10,474	0	1,842	2,842	8,155	13,213	0	2,535	5,367	13,606	23,687
Other ASEAN-9	5,004	3,258	2,727	5,243	5,559	1,359	970	1,872	7,368	5,407	6,362	4,228	4,599	12,611	10,966
China	18,139	27,329	62,760	148,955	194,931	19,505	42,480	53,809	132,163	165,718	37,644	69,809	116,569	281,118	360,649
Other East Asia	2,577	2,648	4,275	7,096	7,056	2,880	3,433	6,517	7,863	7,628	5,458	6,081	10,792	14,959	14,685

(b) Shares in world trade (%)

World	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
United States	12.0	11.4	11.6	11.5	12.2	13.4	18.3	14.7	15.0	18.0	12.7	14.9	13.2	13.3	15.2
European Union-15	41.0	37.8	44.0	39.8	39.0	44.0	36.7	43.8	37.3	36.9	42.5	37.2	43.9	38.5	37.9
East Asia	15.3	19.8	21.2	26.3	25.2	15.2	16.4	19.0	24.7	20.7	15.2	18.0	20.1	25.5	22.9
Japan	7.1	9.5	8.5	8.7	7.4	7.4	6.6	6.7	6.5	5.3	7.2	8.0	7.6	7.6	6.4
Emerging East Asia	8.1	10.3	12.7	17.5	17.8	7.9	9.8	12.3	18.1	15.4	8.0	10.0	12.5	17.8	16.6
NIE-4	4.2	6.1	8.0	10.4	9.8	4.6	5.4	7.7	10.8	9.0	4.4	5.7	7.8	10.6	9.4
Hong Kong SAR	1.1	1.6	2.4	3.4	3.1	1.2	1.5	2.3	3.8	3.1	1.1	1.6	2.4	3.6	3.1
Korea	1.0	1.6	2.0	2.5	2.5	1.1	1.6	2.1	2.6	2.1	1.1	1.6	2.1	2.6	2.3
Singapore	1.1	1.2	1.6	2.3	2.0	1.3	1.3	1.7	2.4	1.9	1.2	1.3	1.6	2.4	2.0
Taiwan	1.1	1.6	2.0	2.2	2.1	1.0	1.0	1.5	2.0	1.9	1.1	1.3	1.7	2.1	2.0
ASEAN-9	2.8	2.7	2.7	4.0	4.5	2.1	2.0	2.9	4.6	3.4	2.5	2.3	2.8	4.3	3.9
Indonesia	1.2	1.0	0.8	0.9	1.0	0.6	0.5	0.6	0.8	0.5	0.9	0.7	0.7	0.8	0.8
Malaysia	0.7	0.8	0.9	1.5	1.5	0.6	0.6	0.8	1.5	1.1	0.6	0.7	0.8	1.5	1.3
Philippines	0.3	0.2	0.2	0.3	0.6	0.4	0.3	0.4	0.6	0.5	0.4	0.3	0.3	0.4	0.6
Thailand	0.4	0.4	0.7	1.1	1.1	0.5	0.5	0.9	1.4	0.9	0.4	0.4	0.8	1.3	1.0
Vietnam	0.0	0.0	0.1	0.1	0.2	0.0	0.1	0.1	0.2	0.2	0.0	0.1	0.1	0.1	0.2
Other ASEAN-9	0.3	0.2	0.1	0.1	0.1	0.1	0.0	0.1	0.1	0.1	0.2	0.1	0.1	0.1	0.1
China	1.0	1.5	1.9	2.9	3.4	1.0	2.1	1.5	2.6	2.9	1.0	1.8	1.7	2.8	3.1
Other East Asia	0.1	0.1	0.1	0.1	0.1	0.2	0.2	0.2	0.2	0.1	0.1	0.2	0.2	0.1	0.1

Source: International Monetary Fund, *International Financial Statistics*, CD-ROM.

Figure 1. **Exports of Developing East Asia and Latin America** (millions of dollars)



Foreign Direct Investment

Foreign direct investment inflows to emerging East Asian economies grew at a remarkably high rate from the mid-1980s to 1998, significantly faster than exports. Indeed, FDI inflows increased more than twelve times in the twelve years from 1985 to 1999 (Table 2 and Figure 2). Unlike exports, FDI inflows continued to grow throughout the period. As result of this rapid expansion, the share of emerging East Asian economies in world FDI inflows increased from 8 percent in 1985 to 22 percent in 1994, before declining sharply to 9 percent in 1999. China increased its share in world FDI inflows from 3 percent in 1985 to 14 percent in 1994, though the share declined to 4 percent in 1999. Despite this decline, China in 1999 was the largest recipient of FDI among emerging market economies and the third largest recipient in the world, behind the United States and the United Kingdom. Compared with emerging East Asian economies, Japan received a very small amount of FDI inflows throughout the period, although it has begun to register a sizable amount in recent years. East Asia increased its share in world FDI inflows in the precrisis period up to 1997, with the share increasing from 7.6 percent in 1985 to 19 percent in 1996 before declining to 10 percent in 1999. The share of East Asia in world FDI inflows is significantly smaller when compared to the case in foreign trade. Indeed, the corresponding shares of the United States and the EU in world FDI inflows are much larger at 30 percent and 40 percent, respectively, in 1999.

Two developments are important concerning recent FDI in emerging East Asia. One is its resilience even during the period of economic crisis. Compared with other forms of international capital flows such as bank lending, which declined precipitously before and after the crisis, FDI inflows remained relatively stable in emerging East Asia, even in those economies that were seriously affected by the crisis. Another important development is the increase in mergers and acquisitions (M&As) as a mode of entry, particularly after the economic crisis (United Nations 1999). Historically, greenfield operations had been the preferred mode of entry for multinationals in East Asia, mainly because of restrictions on equity participation. The economic crisis changed this. Emerging East Asian economies with a keen interest in attracting FDI relaxed those restrictions. Coupled with relaxation of the FDI regime, the huge decline in the values of East Asian currencies and assets encouraged multinationals to undertake M&A.

Turning to FDI outflows from East Asia, one finds that East Asian economies with the exception of Japan and Hong Kong have not been large investors. Although Hong Kong registers sizable FDI outflows, it may not be considered a large investor because a substantial amount of the FDI outflow from Hong Kong originates in other countries. One finds a dramatic decline in the position of Japan as an FDI supplier in the 1990s. In 1990 Japan was the world's largest FDI supplier, supplying 20 percent of world FDI outflows. However, in ten years its share declined to less than 3 percent.

Table 2

Foreign Direct Investment (FDI) Flows of East Asian Economies in the 1980s and 1990s

(a) Millions of U.S. dollars

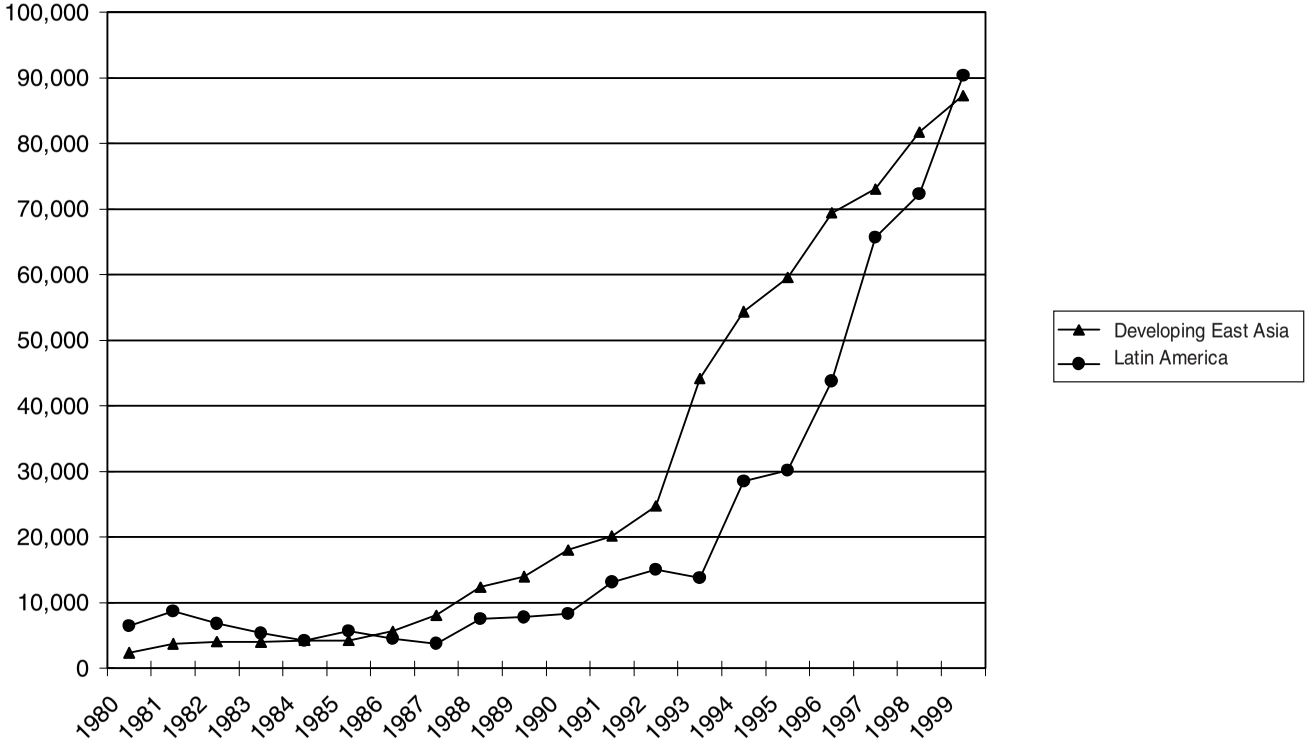
	Inflows					Outflows					Total FDI (inflows + outflows)				
Regions/ countries	1980	1985	1990	1995	19-99	1980	1985	1990	1995	1999	1980	1985	1990	1995	1999
World	52,216	55,502	202,635	318,883	917,481	55,390	59,971	245,820	335,501	871,010	107,606	115,473	448,455	654,384	1,788,491
United States	16,930	20,010	48,497	57,776	275,535	19,230	14,060	37,184	98,750	150,900	36,160	34,070	85,681	156,526	426,435
European Union-15	21,264	15,836	96,173	115,316	371,839	27,067	25,689	130,595	160,931	592,633	48,331	41,525	226,768	276,247	964,472
East Asia	2,654	4,868	19,795	59,590	99,633	2,535	7,964	59,815	39,209	56,918	5,188	12,832	79,609	98,799	156,551
Japan	280	638	1,777	39	12,308	2,390	6,492	50,497	22,508	22,267	2,670	7,130	52,274	22,547	34,575
Emerging East Asia	2,374	4,230	18,017	59,551	87,324	145	1,472	9,318	16,701	34,652	2,518	5,702	27,335	76,252	121,976
NIE-4	1,242	1,280	7,693	10,541	42,312	124	829	8,334	12,816	32,465	1,365	2,109	16,027	23,358	74,777
Hong Kong SAR	0	0	0	0	23,068	0	0	0	0	19,904	0	0	0	0	42,973
Korea	6	234	788	1,776	9,333	26	591	1,052	3,552	4,198	32	825	1,840	5,328	13,531
Singapore	1,236	1,047	5,575	7,206	6,984	98	238	2,034	6,281	3,943	1,333	1,284	7,609	13,488	10,927
Taiwan	1,330	1,559	2,926	5,249	2,983	4,420	0	0	6,579	4,542	7,346				
ASEAN-9	1,018	1,180	6,566	12,593	5,936	3	1	140	1,888	359	1,021	1,181	6,706	14,481	6,294
Indonesia	0	310	1,093	4,346	-2,745	0	0	0	603	72	0	310	1,093	4,949	-2,673
Malaysia	934	695	2,332	4,178	1,553	0	0	0	0	0	934	695	2,332	4,178	1,553
Philippines	–106	12	530	1,478	573	0	0	0	399	–59	–106	12	530	1,877	514
Thailand	190	163	2,444	2,068	6,213	3	1	140	886	346	193	164	2,584	2,954	6,559
Other ASEAN-9	0	0	167	523	342	0	0	0	0	0	0	0	167	523	342
China	0	1,659	3,487	35,849	38,753	0	629	830	2,000	1,775	0	2,288	4,317	37,849	40,528
Other East Asia	114	111	271	567	324	18	14	13	–3	53	132	125	285	564	377

(b) Shares in world FDI (%)

World	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
United States	32.4	36.1	23.9	18.1	30.0	34.7	23.4	15.1	29.4	17.3	33.6	29.5	19.1	23.9	23.8
European Union-15	40.7	28.5	47.5	36.2	40.5	48.9	42.8	53.1	48.0	68.0	44.9	36.0	50.6	42.2	53.9
East Asia	5.1	8.8	9.8	18.7	10.9	4.6	13.3	24.3	11.7	6.5	4.8	11.1	17.8	15.1	8.8
Japan	0.5	1.1	0.9	0.0	1.3	4.3	10.8	20.5	6.7	2.6	2.5	6.2	11.7	3.4	1.9
Emerging East Asia	4.5	7.6	8.9	18.7	9.5	0.3	2.5	3.8	5.0	4.0	2.3	4.9	6.1	11.7	6.8
NIE-4	2.4	2.3	3.8	3.3	4.6	0.2	1.4	3.4	3.8	3.7	1.3	1.8	3.6	3.6	4.2
Hong Kong SAR	0.0	0.0	0.0	0.0	2.5	0.0	0.0	0.0	0.0	2.3	0.0	0.0	0.0	0.0	2.4
Korea	0.0	0.4	0.4	0.6	1.0	0.0	1.0	0.4	1.1	0.5	0.0	0.7	0.4	0.8	0.8
Singapore	2.4	1.9	2.8	2.3	0.8	0.2	0.4	0.8	1.9	0.5	1.2	1.1	1.7	2.1	0.6
Taiwan	0.0	0.0	0.7	0.5	0.3	0.0	0.0	2.1	0.9	0.5	0.0	0.0	1.5	0.7	0.4
ASEAN-9	1.9	2.1	3.2	3.9	0.6	0.0	0.0	0.1	0.6	0.0	0.9	1.0	1.5	2.2	0.4
Indonesia	0.0	0.6	0.5	1.4	-0.3	0.0	0.0	0.0	0.2	0.0	0.0	0.3	0.2	0.8	-0.1
Malaysia	1.8	1.3	1.2	1.3	0.2	0.0	0.0	0.0	0.0	0.0	0.9	0.6	0.5	0.6	0.1
Philippines	–0.2	0.0	0.3	0.5	0.1	0.0	0.0	0.0	0.1	0.0	–0.1	0.0	0.1	0.3	0.0
Thailand	0.4	0.3	1.2	0.6	0.7	0.0	0.0	0.1	0.3	0.0	0.2	0.1	0.6	0.5	0.4
Other ASEAN-9	0.0	0.0	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
China	0.0	3.0	1.7	11.2	4.2	0.0	1.0	0.3	0.6	0.2	0.0	2.0	1.0	5.8	2.3
Other East Asia	0.2	0.2	0.1	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1	0.1	0.1	0.0

Source: International Monetary Fund, *International Financial Statistics*, CD-ROM.

Figure 2. **Foreign Direct Investment Inflows in Developing East Asia and Latin America** (millions of dollars)



Rapid Globalization of Economic Activities in East Asia

Foreign trade and FDI inflows became increasingly important in the economies of emerging East Asia. All of these economies except China and Indonesia registered a ratio of exports to gross domestic products (GDP) exceeding 30 percent, significantly higher than the average ratio of approximately 23 percent for developing economies in 1997 (World Bank 2000). Hong Kong and Singapore had extremely high ratios, which are attributable to their engagement in entrepôt trade. Although high, the ratios for Korea and Taiwan declined from the mid-1980s to the mid-1990s because of the rapid increase in GDP. ASEAN-4 countries, consisting of Indonesia, Malaysia, Philippines, and Thailand, exhibited an increase in the ratio of exports to GDP, reflecting faster growth of exports compared with GDP. Considering that large countries tend to be less dependent on foreign trade than small countries, it is notable that the export to GDP ratios for China and Indonesia exceeded 20 percent.

Although increasing steadily for most developing East Asian economies, the ratio of FDI inflows to GDP was significantly smaller than the ratio of exports to GDP and varied widely within the region. Singapore had the highest ratio, around 10 percent, while Korea and Taiwan had the lowest, around 0.5–1.0 percent. China and Malaysia registered a rapid increase in the ratio, each reaching about 5 percent in 1997. The role of FDI in emerging East Asia's economic activities is more important than the statistics might indicate because the foreign affiliates of multinationals engage actively in production, employment, purchases, and sales, including foreign trade, in the FDI recipient economies.

Factors Behind the Rapid Expansion of Trade and Foreign Direct Investment

The factors behind the significant expansion in foreign trade and FDI inflows in East Asia fall into two groups, one domestic and the other external. The most important domestic factor was the liberalization of both trade and FDI policies. In addition, a favorable macroeconomic environment, reflected in relatively stable price levels, together with an abundant supply of well-educated, low-wage labor contributed to the expansion of exports and FDI inflows. As for external factors, the substantial realignment of exchange rates, particularly the yen-dollar exchange rate in the mid-1980s, was important in promoting exports and FDI inflows. In the 1990s the record-breaking long economic boom in the United States provided huge markets for East Asian exports. In addition, the remarkable technical progress achieved in information technology, which reduced the cost of communications, facilitated global operations by multinational firms. Finally, increased competition among multinational firms, which resulted partly from liberalization and deregulation in various sectors in many countries of the world, promoted their global activities, thereby expanding trade and FDI.

Emerging East Asian economies embarked on liberalization of trade and FDI

policies and deregulation in domestic economic activities as part of more comprehensive structural reform policies. Such policy changes were due to the realization that liberalization and deregulation would promote economic growth. The liberalization of trade and FDI policies led to the expansion of exports and inward FDI because it shifted the incentives from import-substituting production to export production and increased the attractiveness of these economies to foreign investors.

Emerging East Asian economies liberalized their import regimes by lowering tariff rates and nontariff barriers from the early 1980s through the early 1990s (PECC 1995). The notable exceptions were Hong Kong and Singapore, which had adopted virtually free trade regimes. China and Indonesia significantly reduced their average tariff rates. The incidence of nontariff barriers declined in many East Asian economies, except in China. The most remarkable is Indonesia, which reduced nontariff barriers from 95 percent in 1984–87 to less than 3 percent in 1991–93. Korea also reduced both its tariff rates and the incidence of nontariff barriers during the period from 1988 to 1993.

Inward FDI policies were liberalized in the mid-1980s, as East Asian economies began to realize that FDI inflows would promote economic growth. Although it is difficult to quantify the restrictiveness of an FDI regime, it is clear that many emerging East Asian economies have liberalized their FDI policies since the mid-1980s (Yamazawa and Urata 2001). Restrictions on FDI took various forms, including restrictions on market access, most-favored-nation treatment, and national treatment. Many emerging East Asian economies reduced the restrictions on market access by reducing the number of sectors and industries on the negative list and by relaxing the limits on foreign equity ownership. Among emerging East Asian economies, Indonesia, Korea, Malaysia, the Philippines, and Thailand adopted substantial FDI liberalization measures in an effort to attract foreign investors. Furthermore, recognizing the important contribution that FDI may make toward economic growth, a number of economies introduced incentives such as tax breaks to attract FDI. Indeed, there has been keen competition among emerging East Asian economies to attract FDI by reducing barriers and providing incentives.

Liberalization of trade and FDI also progressed under regional and global frameworks. The members of ASEAN formed the ASEAN Free Trade Area (AFTA) in 1992, the only formal regional trade arrangement in East Asia. The 1992 agreement provided for the liberalization of tariff and nontariff measures under the Common Effective Preferential Tariffs. The target year for achieving tariff and nontariff liberalization was originally set for 2008, but was later moved forward to 2003. FDI liberalization in the ASEAN has been under way since the creation in 1998 of the ASEAN Investment Area (AIA), which provides coordinated investment cooperation and facilitation programs, market access, and national treatment of all industries. The target dates for the creation of AFTA were 2003 for the original ASEAN members and Myanmar (which did, for the most part, meet the target) and 2010 for Vietnam, Laos, and Cambodia.

Asia-Pacific Economic Cooperation (APEC) has also contributed to the liberal-

ization and facilitation of trade and FDI for emerging East Asian economies. This transregional forum includes not only East Asian economies but also countries in North and South America and Oceania. Following the Bogor declaration in 1994 calling for full liberalization of trade and FDI by 2010 for developed-country members and by 2020 for developing-country members, APEC members agreed to prepare and implement individual action plans specifying near- and medium-term liberalization measures. Peer pressure is expected to play a crucial role in implementation. All APEC members have made significant progress toward freer trade and FDI regimes.

The Uruguay Round of multilateral trade negotiations under the GATT started in 1986 and ended in 1994. Although the negotiations lasted eight years, the Uruguay Round made substantial progress toward liberalizing trade and FDI. The achievements include: a reduction in tariff rates, framework agreements on trade in services, on intellectual property rights, and on trade-related investment measures, a timetable for phasing out all quantitative restrictions on trade, first steps toward bringing agriculture more firmly under a multilateral discipline, a stronger dispute settlement mechanism, and the establishment of the WTO. Though it is difficult to estimate the impact of these achievements individually, there is no doubt that the GATT/WTO has promoted trade and FDI liberalization in East Asia.

Increasing Intraregional Dependence on Trade and Foreign Direct Investment in East Asia

The preceding section reviewed the recent developments in East Asia's trade[2] and FDI vis-à-vis the rest of the world. In other words, it focused on the globalization of East Asia. This section turns to the subject of regionalization in East Asia by reviewing trade and FDI patterns. The analysis reveals the formation of a regional production system by multinational corporations, which appears to have contributed to regionalization in East Asia.

The Measurement of Intraregional Dependence

Several studies have examined the changes in intraregional dependence in foreign trade in East Asia. Computing three sets of measures, Petri (1993) finds that intraregional dependence in foreign trade in East Asia increased steadily in the post–World War II period, after declining in the pre–World War II period, and that intraregional bias declined in the post–World War II period. Frankel (1993) also finds a decline in intraregional bias in foreign trade in the 1980s by estimating the magnitude of the bias in the gravity model framework. This subsection investigates the changing patterns of intraregional trade and FDI in East Asia from the early 1980s to the late 1990s. Following Petri (1993), three measures are computed for foreign trade and FDI: absolute measures, relative measures, and double-relative measures.

Absolute measure (A): $A = Xij / X..$
Relative measure (B): $B = A / (Xi. / X..) = Xij / Xi.$
Double-relative measure (C): $C = A / [(Xi. / X..) (X.j / X..)] = Xij * X.. / Xi. * X.j$,

where Xij represents exports (or outward FDI) from region i to region j, and "." indicates the summation across all i or j. Therefore, $Xi.$ represents total exports (or outward FDI flows) of region i, $X.j$ represents total imports (or inward FDI flows) of region j, and $X..$ represents world trade (or world total FDI flows).

The absolute measure compares the scale of a particular bilateral trading (or FDI) relationship to world trade (or world FDI), while the relative measure compares it to trade (or FDI) of one or the other of the two partners participating in the relationship. The double-relative measure, which is commonly called the intensity index, shows the intensity or bias of the bilateral trading (or FDI) relationship by taking into account its importance in world trade (or world FDI). The value of unity for the double-relative measure can be interpreted so that the bilateral relationship is neutral, while the relationship is more (or less) biased when the measure is greater (or less) than unity.

Foreign Trade

Table 3 shows the estimated values of the three measures of foreign trade and FDI for several regions in the world—East Asia, the North American Free Trade Agreement (NAFTA) region, the European Union, and the MERCOSUR region. The results indicate that intraregional trade in East Asia became more important not only in terms of world trade but also in terms of regional trade. However, intraregional bias became smaller over time.

The importance of intra–East Asian trade in world trade increased significantly from 5 percent in 1980 to 13 percent in 1995, though it declined slightly to 11 percent in 1999. The share of intra-NAFTA trade in world trade also increased over the same period, but the share was smaller, at 10 percent in 1999. The corresponding share for the European Union was significantly higher, at 23 percent in 1999, although the share had declined sharply from 29 percent in 1990.

A significant part of intra–East Asian trade takes place between Japan and emerging East Asian economies. This can be seen from the fact that the share of intraregional trade among emerging East Asian economies (6 percent in 1999) amounts to only slightly more than half of the level observed for East Asia as a whole (11 percent), that is, including Japan. The magnitude of intraregional trade for the Asian NIE-4 and for the ASEAN-10 is still quite small in world trade, amounting in 1999 to 1.2 percent and 1.4 percent, respectively, of world trade.

Intra–East Asian trade increased its importance for East Asia's total trade (exports plus imports) over time, as shown in the increase in the relative measure from 36 percent in 1980 to 50 percent in 1997, followed by a slight decline to 49 percent in 1999. The comparable figures for emerging East Asia were 22 percent

and 39 percent in 1980 and 1999, respectively. The importance of intraregional trade in total regional trade also increased for the members of NAFTA from 33 percent to 47 percent during the same period, but it declined for the European Union from 65 percent in 1990 to 61 percent in 1999, after rising between 1980 and 1990. Among the subgroups in East Asia, intragroup trade among the NIE-4 was quite small, amounting to only 13 percent of total trade, while intra–ASEAN-10 trade was larger, at 23 percent.

An analysis of the relative measures computed for exports and for imports shows that intra–East Asian trade is more important as a source of imports than as a destination for exports. This finding indicates a trading pattern in which East Asian economies procure imports within the region and sell exports outside the region. This appears to reflect the behavior of multinationals, as will be confirmed in a later section. Many multinationals use East Asia as an export platform, in which they assemble export products for regions outside of East Asia by importing parts and components from within the region. In contrast, intra-NAFTA trade is more important for NAFTA's exports than for its imports.

The results of the double-relative measure (or the trade intensity index) reveal an interesting contrast concerning the intraregional trade bias for East Asia, on the one hand, and the NAFTA and the EU, on the other hand. Specifically, intraregional bias declined in East Asia from 2.6 in 1980 to 2.3 in 1999, while the corresponding values for the NAFTA and the EU increased from 2.1 and 1.5 to 2.3 and 1.7, respectively, over the same period. Among East Asian subgroups, intraregional trade bias is very high for ASEAN, with the double-relative measure at 4.2 in 1999, although the size of the bias has declined over time.

The estimated measures of intraregional trade dependence reveal that the importance of intraregional trade in East Asia increased not only in world trade but also in regional trade over time. However, extraregional trade also expanded rapidly. Indeed, intraregional trade bias declined in East Asia, while it increased in the NAFTA and the EU. One may attribute these differences partly to differences in the institutional arrangements. Both the NAFTA and the EU have trade arrangements that give preferential treatment to their members, possibly leading to an increasing regional bias. In East Asia, a preferential trading arrangement has been set up only for the ASEAN members that make up a small portion of intra–East Asian trade, and other economies do not have any discriminatory arrangements. The absence of discriminatory trade arrangements may have caused a decline in trade bias. Unilateral trade liberalization without discriminatory treatment among trading partners, even including those of ASEAN, may have contributed to a decline in regional trade bias in East Asia. Furthermore, a decline in the cost of communications and transportation services, resulting from technological progress and liberalization, contributed to the diversification of trading partners. Rapid industrialization centered on similar industries, such as textiles and electric machinery, has forced many East Asian economies to look outside the region for markets for their products, diminishing the intraregional trade bias (Petri 1993).

Table 3

Intrareligious Dependence for Trade and Foreign Direct Investment (FDI)

(a) Absolute measure (%)

	Trade					FDI	
	1980	1985	1990	1995	1999	1980	1994
East Asia, including Japan	5.4	6.7	8.3	12.7	11.3	3.5	8.4
Emerging East Asia-14	1.7	2.5	3.9	6.6	6.4	1.5	5.5
NIE-4	0.4	0.5	0.9	1.5	1.2	0.1	0.1
ASEAN-10	0.7	0.7	0.8	1.5	1.4	0.5	0.8
ASEAN-9, excluding Singapore	0.1	0.1	0.1	0.3	0.4	—	—
NAFTA	5.7	7.9	6.6	7.8	10.3	13.6	5.3
MERCOSUR	0.2	0.1	0.1	0.3	0.3	—	—
European Union-15	24.3	21.7	28.3	23.8	23.1	12.9	19.3

(b) Relative measure (%)

	Exports					Outward FDI	
East Asia, including Japan	35.9	34.5	40.1	49.3	45.0	38.1	35.0
Emerging East Asia-14	23.3	26.4	32.4	39.6	37.0	86.8	86.6
NIE-4	9.7	8.9	12.4	16.1	14.8	8.6	2.8
ASEAN-10	18.7	19.8	19.8	25.4	22.2	73.2	56.5
ASEAN-9, excluding Singapore	3.9	5.1	5.0	7.3	8.1	—	—
NAFTA	33.6	43.9	41.4	46.2	54.6	27.5	18.7
MERCOSUR	11.6	5.5	8.9	20.3	20.5		
European Union-15	60.8	59.2	65.9	62.4	62.8	38.6	47.9
	Imports					Inward FDI	
East Asia, including Japan	35.3	40.5	43.6	51.2	54.1	42.0	54.0
Emerging East Asia-14	21.1	24.2	30.6	34.6	40.5	19.7	38.3

NIE-4	7.1	8.2	10.3	11.8	11.4	4.1	4.6
ASEAN-10	18.2	19.7	16.4	19.4	23.9	10.2	11.3
ASEAN-9, excluding Singapore	5.0	6.9	4.8	7.3	10.7	—	—
NAFTA	32.8	34.4	33.9	38.4	41.1	41.5	21.2
MERCOSUR	8.3	9.5	14.2	18.2	19.2	—	—
European Union-15	54.0	57.5	63.1	61.3	59.6	36.2	55.0
	Trade (exports + imports)					FDI (out + in)	
East Asia, incl. Japan	35.6	37.3	41.8	50.2	49.2	40.0	42.5
Emerging East Asia-14	22.2	25.3	31.5	37.1	38.7	32.1	53.1
NIE-4	8.3	8.6	11.3	13.9	13.2	5.6	3.5
ASEAN-10	18.4	19.7	18.0	22.2	22.9	17.9	18.8
ASEAN-9, excl. Singapore	4.3	5.9	4.9	7.3	9.2	—	—
NAFTA	33.2	38.3	37.2	42.0	46.8	33.0	19.9
MERCOSUR	9.7	7.0	11.0	19.2	19.8	—	—
European Union-15	57.2	58.3	64.5	61.9	61.2	37.3	51.2
(c) Double-relative measure (trade intensity index, %)							
	Trade					FDI	
East Asia, including Japan	2.56	2.28	2.27	2.14	2.25	4.59	2.25
Emerging East Asia-14	2.97	2.93	2.84	2.32	2.49	11.36	6.08
NIE-4	2.09	1.75	1.70	1.64	1.75	4.20	0.98
ASEAN-10	5.12	6.06	4.60	3.80	4.15	14.09	7.59
ASEAN-9, excluding Singapore	1.70	2.65	1.87	1.64	2.34	—	—
NAFTA	2.05	1.96	2.15	2.39	2.27	0.84	0.74
MERCOSUR	6.55	4.93	9.70	13.27	14.68	—	—
European Union-15	1.48	1.60	1.51	1.66	1.66	1.08	1.37

Sources: Trade data: International Monetary Fund, Direction of Trade Statistics; FDI data: Industry Canada.
Note: Definitions are described in text.

Foreign Direct Investment

Similar to the changing patterns of trade, intraregional FDI in East Asia increased from 4 percent of world FDI in 1980 to 8 percent in 1994. The corresponding share for the EU also increased from 13 percent to 19 percent, while it declined for the NAFTA from 14 percent to 5 percent. Among the East Asian subgroups, the stock of intraregional FDI in the emerging economies registered relatively high growth, increasing from 2 percent to 6 percent of world FDI during the 1980–94 period.

Intraregional FDI in East Asia increased from 40 percent in 1980 to 43 percent in 1994. Among the subgroups, intraregional FDI became particularly important for emerging East Asia. Coupled with this observation, the relatively small shares of intraregional FDI for NIEs and ASEAN indicate the importance of FDI from the former to the latter subgroup. Intraregional FDI is particularly important because 87 percent of outward FDI has been directed to emerging East Asia. The share of inward FDI has increased in East Asia, which means that an increasing share of inward FDI originates inside the region. However, the share of intraregional FDI in regional FDI is substantially smaller for emerging East Asian economies, reflecting the importance of Japan as a source of FDI.

The results of double-relative measures show an interesting contrast between East Asia and the EU. Although the magnitude of the bias is higher for East Asia than for the EU, the magnitude of the bias declined for East Asia, while it increased for the EU. The extent of the bias remained more or less the same for the NAFTA. These observations are consistent with those made for foreign trade, and differences in the direction of bias for East Asia and for the EU may reflect differences in institutional arrangements, as argued in the case of foreign trade.

Patterns similar to those of trade are found for intraregional FDI in East Asia. That is, from 1980 to 1994 the importance of intraregional FDI in East Asia increased not only in world FDI but also in overall regional FDI in East Asia. However, a regional bias declined during the 1980–94 period. These findings indicate that the increasing importance of intraregional trade and FDI is attributable largely to the rapid expansion of overall trade and FDI in the region that is driven by market forces. This contrasts with the case in the EU or the NAFTA, where intraregional bias in foreign trade and FDI increased possibly because of discriminatory institutional arrangements under which regional members receive preferential treatment, worsening resource allocation.

A lack of necessary information precludes us from computing the measures for intraregional dependence in FDI for the period after 1994. However, some evidence indicates a declining intraregional dependence in FDI for East Asia since the mid-1990s. One piece of evidence is substantial declines in the shares of East Asia in world FDI inflows and outflows in the latter half of the 1990s (Table 2). Another is the sharp decline in the importance of East Asian economies as foreign investors in East Asia. According to the figures in Table 4, East Asia lost its share

as an FDI supplier in all the East Asian economies listed in the table except Indonesia. In contrast to the position of East Asia, the EU increased its share in many East Asian economies. These findings indicate the declining importance of East Asia not only in world FDI but also as a supplier of FDI in East Asia, which in turn tend to show a decline in intraregional dependence in FDI.

Creation of an Intraregional Production System in East Asia

Increasing Intraregional Dependence in Production

So far the changing patterns of foreign trade and FDI in East Asia have been examined without considering their relationship with the economic structures of the East Asian economies. Our earlier observation that foreign trade of the East Asian economies and in particular intraregional trade in East Asia expanded rapidly since the 1980s may indicate a substantial impact on the economic structure of the East Asian region as well as those of the individual economies. This section investigates intraregional trade patterns in East Asia by explicitly relating them to procurement sources of the inputs for production by using international input-output tables. From the analysis, the changing characteristics of intraregional, interindustry relationships in East Asia will be discerned.

The Institute of Developing Economies in Japan constructed international input-output tables covering East Asian economies and the United States for 1985, 1990, and 1995. The international input-output tables are constructed by linking input-output tables of the individual economies by explicitly specifying the import sources and export destinations of the products. The international input-output table shows the source of inputs for production, that is, inputs from the domestic market and imported inputs from other economies. Like input-output tables, the international input-output table shows the destination of outputs, that is, outputs sold in the domestic market and those exported to other economies. Below we examine the source of inputs for production by East Asian economies, because our main interest here is to examine the interindustry, intraregional production relationship in East Asia.

Table 5 shows the production (input) structures of East Asian economies in 1985 and 1995. Between 1985 and 1995 the importance of imported inputs in production increased for the East Asian economies except Singapore, Korea, and Japan, indicating increasing dependence on foreign countries for the supply of intermediate inputs. Wide variations in the importance of imported inputs in production can be observed for East Asian economies. Malaysia, the Philippines, Singapore, Thailand, and Taiwan exhibit high dependence as the share of imported inputs in production for these economies exceeded 10 percent. Indonesia, China, and Korea have low shares of around 6 percent.

To examine intraregional dependence in production in East Asia, we computed the share of imported inputs, which originated inside the region, in total inputs

Table 4

Sources of Inward Foreign Direct Investment in East Asian Economies (%)

	China (actualized value)			Indonesia (approved)			Korea (approved)			Malaysia (approved)		
	1990	1995	1999	1990	1995	1999	1990	1995	1999	1991	1995	1999
World	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
United States	13.1	8.2	10.5	1.8	6.9	1.3	39.6	33.2	24.1	10.5	19.7	42.0
East Asia	69.8	77.9	64.1	55.3	20.8	30.0	31.9	29.0	17.3	65.9	58.5	18.6
Japan	14.4	8.3	7.4	25.6	9.5	5.9	29.3	21.5	11.3	21.7	22.9	8.2
Korea	0.0	2.8	3.2	8.3	1.7	2.4	0.0	0.0	0.0	10.7	6.6	0.3
Taiwan	0.0	8.4	6.4	7.1	1.4	13.7	0.5	0.5	0.2	21.1	15.8	2.2
Hong Kong	53.9	53.5	40.6	11.4	4.4	0.7	0.4	3.0	3.0	3.5	1.9	0.5
Singapore	1.4	4.9	6.6	3.0	3.7	6.7	1.7	3.4	2.7	6.5	11.0	7.3
China	0.0	0.0	0.0	0.0	0.0	0.5	0.0	0.6	0.2	2.3	0.2	0.1
European Union	3.4	5.2	10.0	8.1	21.4	6.1	20.8	17.5	35.5	4.7	5.6	10.0
Italy	0.1	0.7	0.5	0.0	0.1	0.0	0.2	0.2	0.0	0.2	0.2	0.6
United Kingdom	0.4	2.4	2.6	0.7	15.8	4.7	5.6	4.5	3.1	3.2	2.1	1.6
Netherlands	0.5	0.3	1.3	6.5	0.9	0.4	4.5	8.8	21.4	0.0	0.6	6.3
Germany	1.8	1.0	3.4	0.2	3.4	0.8	7.8	2.3	6.2	1.1	1.6	1.5
France	0.6	0.8	2.2	0.8	1.2	0.2	2.8	1.8	4.9	0.2	1.1	0.1

	Philippines (BOP)			Singapore (committed)			Taiwan (approved)			Thailand (BOP)		
	1990	1995	1999	1990	1995	1999	1990	1995	1999	1990	1995	1999
World	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0	100.0
United States	26.9	6.8	4.5	47.6	42.8	57.3	25.3	44.6	222.7	9.5	13.0	8.0
East Asia	46.6	70.1	26.4	31.9	23.8	18.9	50.0	27.3	25.3	75.5	54.0	46.4
Japan	27.7	30.0	16.0	31.9	23.8	18.9	36.4	19.6	12.2	43.2	27.8	12.2
Korea	3.9	1.0	0.7	0.0	0.0	0.0	0.1	0.1	0.1	0.8	0.6	0.2
Taiwan	3.9	0.9	0.8	0.0	0.0	0.0	0.0	0.0	0.0	11.1	4.8	3.9
Hong Kong	7.9	28.9	1.1	0.0	0.0	0.0	10.3	5.0	3.8	10.9	13.9	7.2
Singapore	3.2	9.3	1.9	0.0	0.0	0.0	3.2	2.6	9.2	9.5	6.8	23.0
China	0.0	0.0	5.9	0.0	0.0	0.0	0.0	0.0	0.0	0.2	0.1	-0.1
European Union	7.7	12.9	22.7	17.5	30.9	17.4	8.4	10.4	9.4	5.7	12.3	41.6
Italy	0.0	0.0	0.0	0.0	0.3	1.4	0.0	0.0	0.0	0.1	-0.3	0.0
United Kingdom	4.4	6.5	0.5	4.1	15.9	1.5	3.9	6.7	3.4	1.7	2.8	5.1
Netherlands	1.7	3.7	20.3	3.3	8.1	3.0	2.2	2.6	4.6	1.0	4.3	18.1
Germany	1.1	2.0	1.3	7.5	3.8	10.1	1.6	0.8	0.7	1.8	1.9	8.2
France	0.5	0.8	0.6	2.7	2.9	1.5	0.8	0.4	0.8	1.1	3.6	10.2

Source: Country sources.

Table 5

Sources of Inputs in Production for East Asian Economies, 1985 and 1995

		Indonesia	Malaysia	Philippines	Thailand	Singapore	Taiwan	Korea	China	Japan
Domestic inputs	1985	0.3517	0.3367	0.3843	0.4073	0.2976	0.4614	0.4435	0.5008	0.4737
	1995	0.3920	0.3371	0.3385	0.3719	0.3433	0.3869	0.4782	0.5547	0.4395
Imported inputs	1985	0.0624	0.1718	0.0797	0.0996	0.3353	0.1371	0.1290	0.0417	0.0481
	1995	0.0692	0.2396	0.1452	0.1605	0.2868	0.1621	0.0612	0.0614	0.0306
From NIEs-3	1985	0.0062	0.0298	0.0045	0.0106	0.0117	0.0022	0.0024	0.0010	0.0022
	1995	0.0079	0.0397	0.0198	0.0189	0.0260	0.0090	0.0031	0.0071	0.0022
From ASEAN-4	1985	0.0007	0.0073	0.0067	0.0112	0.0635	0.0062	0.0095	0.0011	0.0053
	1995	0.0018	0.0096	0.0084	0.0097	0.0553	0.0094	0.0060	0.0026	0.0027
China	1985	0.0012	0.0028	0.0040	0.0021	0.0350	0.0000	0.0000	0.0000	0.0019
	1995	0.0018	0.0052	0.0036	0.0042	0.0072	0.0044	0.0053	0.0000	0.0015
From Emerging East Asia	1985	0.0080	0.0399	0.0151	0.0239	0.1102	0.0083	0.0119	0.0021	0.0094
	1995	0.0115	0.0545	0.0317	0.0327	0.0884	0.0228	0.0144	0.0097	0.0064
From Japan	1985	0.0121	0.0287	0.0051	0.0150	0.0338	0.0242	0.0241	0.0095	0.0000
	1995	0.0100	0.0560	0.0191	0.0339	0.0663	0.0342	0.0196	0.0095	0.0000
From East Asia	1985	0.0201	0.0686	0.0202	0.0389	0.1440	0.0325	0.0360	0.0115	0.0094
	1995	0.0216	0.1105	0.0508	0.0666	0.1547	0.0569	0.0340	0.0193	0.0064
From rest of the world	1985	0.0423	0.1033	0.0595	0.0607	0.1913	0.1045	0.0930	0.0302	0.0387
	1995	0.0476	0.1291	0.0944	0.0939	0.1320	0.1051	0.0271	0.0421	0.0242
Total inputs	1985	0.4141	0.5085	0.4640	0.5069	0.6329	0.5985	0.5724	0.5425	0.5218
	1995	0.4612	0.5767	0.4837	0.5324	0.6301	0.5490	0.5393	0.6160	0.4700
Value added	1985	0.5859	0.4915	0.5360	0.4931	0.3671	0.4015	0.4276	0.4575	0.4782
	1995	0.5388	0.4233	0.5163	0.4676	0.3699	0.4510	0.4607	0.3840	0.5300
Output	1985	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
	1995	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Source: Institute of Developing Economies, International Input-Output Tables, 1985 and 1995 versions.

(total intermediate goods and value added). The results show an increase in intraregional dependence in production for East Asian economies except Korea and Japan. Singapore and Malaysia register particularly high dependence, while dependence for China and Japan is quite low.

The foregoing analysis did not reveal "true" intraregional dependence in production, as it only took account of direct inputs and did not consider indirect inputs, which are required for the production of inputs. To shed light on the true intraregional dependence, we computed the magnitude of output being induced by a unit increase in final demand in a particular economy in East Asia by using international input-output tables. The results of the computation, which incorporate not only interindustry relationships inside the economy but also those with other East Asian economies, are presented in Table 6. For example, in 1985 one unit increase in final demand in Indonesia increases the output of Indonesia by 1.6026 units.[3] It also increases the output of NIEs-3 by 0.0172.

The figures in Table 6 show that intraregional, interindustry relationships deepened from 1985 to 1995 because for East Asia a higher level of output is obtained in 1995 (16.2102 units) compared with 1985 (16.1393 units) from a unit increase in final demand of all East Asian economies. Although a large part of induced production is realized in the economy registering an increase in final demand, output of other economies also increases. Indeed, the induced level of output in other East Asian economies excluding output induced in an economy having an increase in final demand increased from 0.9762 in 1985 to 1.3591 in 1995, indicating a "net" deepening in intraregional, interindustry relationships. However, the magnitude of the contribution to the deepening of intraregional, interindustry relationships is not uniform among East Asian economies. Concerning the "net" contribution, all the East Asian economies excluding Indonesia and Japan contributed to the deepening of the relationship. An examination of the magnitude of output induced in East Asian economies shows more or less equal dependence on other emerging East Asian economies and Japan for the supply of intermediate inputs. It should be noted that their dependence on Japan increased significantly between 1985 and 1995, probably indicating that the large amount of FDI from Japan promoted imports of inputs from Japan. This finding also indicates the increasing importance of emerging East Asian economies for Japan's economic growth.

Multinational Firms: Promoters of Regional Economic Integration

East Asian economies have experienced a rapid expansion of foreign trade and FDI in recent years. We observed increasing intraregional dependence on foreign trade and production in East Asia. Recognizing the increasingly important role of FDI in economic activities, we shall investigate the behavior of multinational firms and major foreign investors, to examine their role in regional integration. Specifically, we will analyze the procurement and sales patterns of their affiliates in Asia.

Table 6

Intereconomy, Interindustry Linkages in East Asia

	An increase in final demand in									
	Indonesia	Malaysia	Philippines	Thailand	Singapore	Taiwan	Korea	China	Japan	Total
Increase in production										
Domestic production										
1985	1.6026	1.4722	1.6189	1.7045	1.5308	1.7638	1.6971	1.8965	1.8768	15.1631
1995	1.5680	1.4469	1.4850	1.5685	1.5083	1.6420	1.6553	2.1608	1.8161	14.8511
1985–95	–0.0346	–0.0253	–0.1338	–0.1360	–0.0225	–0.1218	–0.0417	0.2644	–0.0607	–0.3121
NIEs-3										
1985	0.0172	0.0914	0.0141	0.0278	0.0275	0.0050	0.0060	0.0029	0.0067	0.1987
1995	0.0171	0.0728	0.0557	0.0381	0.0567	0.0222	0.0063	0.0227	0.0066	0.2981
1985–95	–0.0001	–0.0187	0.0417	0.0102	0.0292	0.0172	0.0003	0.0197	–0.0001	0.0995
ASEAN-4										
1985	0.0029	0.0186	0.0157	0.0159	0.1106	0.0119	0.0152	0.0031	0.0224	0.2163
1995	0.0040	0.1088	0.0184	0.0162	0.0979	0.0221	0.0222	0.0078	0.0099	0.3072
1985–95	0.0011	0.0901	0.0027	0.0003	–0.0128	0.0102	0.0071	0.0048	–0.0125	0.0909
China										
1985	0.0041	0.0160	0.0130	0.0091	0.0714	0.0008	0.0007	0.0000	0.0061	0.1211
1995	0.0053	0.0135	0.0135	0.0112	0.0281	0.0165	0.0153	0.0000	0.0056	0.1090
1985–95	0.0012	–0.0026	0.0006	0.0021	–0.0433	0.0157	0.0146	0.0000	–0.0005	–0.0121

Emerging East Asia excluding own domestic production										
1985	0.0242	0.1261	0.0428	0.0528	0.2096	0.0177	0.0218	0.0060	0.0352	0.5361
1995	0.0264	0.1950	0.0877	0.0655	0.1827	0.0608	0.0438	0.0305	0.0221	0.7144
1985–95	0.0022	0.0689	0.0449	0.0127	–0.0269	0.0431	0.0219	0.0245	–0.0131	0.1783
Emerging East Asia including own domestic production										
1985	1.6268	1.5983	1.6616	1.7573	1.7404	1.7815	1.7189	1.9025	1.9120	15.6992
1995	1.5944	1.6418	1.5727	1.6340	1.6910	1.7028	1.6991	2.1913	1.8382	15.5654
1985–95	–0.0324	0.0436	–0.0889	–0.1233	–0.0494	–0.0787	–0.0198	0.2889	–0.0738	–0.1338
Japan										
1985	0.0378	0.0839	0.0206	0.0488	0.0897	0.0627	0.0639	0.0327	0.0000	0.4401
1995	0.0263	0.1221	0.0657	0.0769	0.1777	0.0922	0.0479	0.0359	0.0000	0.6448
1985–95	–0.0115	0.0382	0.0451	0.0281	0.0880	0.0296	–0.0160	0.0032	0.0000	0.2047
East Asia excluding own domestic production										
1985	0.0619	0.2100	0.0634	0.1017	0.2993	0.0803	0.0857	0.0387	0.0352	0.9762
1995	0.0527	0.3170	0.1534	0.1424	0.3604	0.1530	0.0916	0.0664	0.0221	1.3591
1985–95	–0.0092	0.1071	0.0900	0.0407	0.0611	0.0727	0.0060	0.0277	–0.0131	0.3830
East Asia including own domestic production										
1985	1.6646	1.6822	1.6822	1.8062	1.8301	1.8441	1.7828	1.9352	1.9120	16.1393
1995	1.6207	1.7639	1.6384	1.7109	1.8688	1.7951	1.7470	2.2272	1.8382	16.2102
1985–95	–0.0439	0.0817	–0.0438	–0.0953	0.0386	–0.0491	–0.0358	0.2920	–0.0738	0.0709

Source: Institute of Developing Economies, International Input-Output Tables, 1985 and 1995 versions.

To examine this issue in a comprehensive way, we need detailed information on trade by multinationals in East Asia. However, necessary data to carry out such an analysis are not available for foreign firms of all origins in East Asia. Below we use the information on Japanese multinationals to examine this issue. Although limited in its coverage, the data on Japanese multinationals will provide us with useful information on foreign firms' trading patterns in East Asia because Japanese multinationals account for a large portion of multinationals operating in East Asia.

Table 7 presents the procurement and sales patterns of Japanese manufacturing multinationals in Asia in 1992 and 1998. One observes that for the Asian affiliates of Japanese firms, the share of local procurements and sales in their total declined and the share of foreign trade increased from 1992 to 1998. Specifically, the shares of imports in total procurements and the shares of exports in total sales for their Asian affiliates increased from 51 percent and 34 percent in 1992 to 56 percent and 50 percent in 1998, respectively. The corresponding figures for the North American affiliates changed from 49 percent and 8 percent to 48 percent and 12 percent, while the figures for European affiliates declined from 71 percent and 44 percent to 59 percent and 40 percent (not shown in the table). These data indicate that the Asian affiliates exhibit a greater trade orientation when compared with those in North America, but they record a lower trade orientation than those in Europe.

An examination of the regional composition of foreign trade for the overseas affiliates of Japanese firms reveals the increasing importance of intraregional trade in Asia. For imports by the Asian affiliates, the share of Japan and the rest of Asia (labeled "Asia" in the table) in total procurements increased from 47 percent in 1992 to 53 percent in 1998. The corresponding shares for exports in total sales increased from 27 percent to 42 percent. It is of interest to observe that for procurements by the Asian affiliates dependence on Asia excluding Japan increased from 1992 to 1997, while dependence on Japan alone declined, possibly reflecting a shift in the sourcing of imports from Japan to the rest of Asia. By contrast, for sales by Asian affiliates, dependence on both Japan and the rest of Asia increased at the expense of other regions' exports. Among manufacturing subsectors, electronic machinery affiliates exhibit high dependence on trade, especially on intraregional trade, in their procurements and sales, reflecting the presence of a regional production network.

A rapid expansion of foreign trade in Asia by Japanese multinationals was attributable to the rapid expansion of intrafirm trade, that is, foreign trade between the Japanese parent and its foreign affiliates, or between foreign affiliates. In particular, intrafirm trade accounts for a substantial share of foreign trade by the Asian affiliates in their trade with Japan. In 1998, as much as 96 percent of those affiliates' exports to Japan were destined to their parents (Table 8). Although dependence on intrafirm trade is less for imports, as much as 83 percent of the Asian affiliates' imports from Japan came from their parents. The importance of intrafirm

trade is lower for the trade between affiliates in Asia, when compared with the case for trade between the affiliates and their parents.

Two interesting observations may be made from the findings on the procurement and sales patterns of Japanese multinational firms. First, Japanese multinationals have contributed to the promotion of foreign trade in Asia, in particular intraregional trade. Second, Japanese multinationals have become more active in pursuing an intraregional, interprocess division of labor, probably contributing to the efficient use of factors of production in Asia.

A large number of multinationals in East Asia have been found to be of the efficiency-seeking type, not the market-seeking type. As such, these multinationals locate themselves in an economy where they can perform their operation most efficiently or at the least cost. Japanese multinationals in machinery sectors such as electronics, which account for a large part of Japanese multinational activity in East Asia, break up their production process into several subprocesses, and locate each subprocess in an economy where that particular subprocess may be carried out most efficiently. For example, some Japanese multinationals manufacturing televisions break up the production process into subprocesses such as parts production and assembly operation, and they locate these subprocesses in economies where the required factor inputs are relatively abundant, for example, high-skilled workers for parts production and low-wage labor for assembly operation. The TV manufacturers export parts to an economy where the final products are assembled, and export the assembled TVs to other economies.

U.S. multinationals also have been active in setting up production networks in East Asia. Unlike the more or less closed production system of Japanese multinationals, production networks constructed by U.S. multinationals are said to be more open to firms from other nationalities such as those from Korea, Singapore, and Taiwan. Indeed, the basic strategy of U.S. multinationals is to link up with the most efficient producers regardless of their nationality.[4] Like Japanese multinationals, U.S. multinationals have contributed to the increased trade orientation of the host East Asian economies, but they have not contributed much to the promotion of intraregional trade, as their ties with the U.S. market are expectedly strong (Dobson 1997).

Many firms from the Asian NIEs also set up production networks in various parts of the world, particularly in East Asia. One of the industries that has actively pursued such a globalization strategy is textiles. All these production systems and networks clearly have contributed to greater intraregional trade interdependence in East Asia.[5]

Recent Surge of Free Trade Agreements in East Asia

The analysis in the previous sections found market forces to be the major driver of regional economic integration in East Asia. This pattern began to change in the late 1990s, as many East Asian countries became interested in institutional

Table 7

Procurements and Sales of Foreign Affiliates of Japanese Firms: 1992 and 1998 (percentage of total procurements or sales)

	Procurements 1992									
	World					Asia				
		imports from					imports from			
	Local procure-ments	Japan	North America	Asia	Europe	Local procure-ments	Japan	North America	Asia	Europe
Manufacturing total	46.5	40.9	1.3	5.2	5.0	48.5	37.9	1.8	8.9	0.5
Food	84.8	6.1	0.1	7.0	0.2	72.0	4.5	0.3	22.9	0.1
Textiles	44.4	20.8	4.0	9.8	6.1	40.7	22.4	4.7	12.1	1.4
Wood and pulp	89.1	7.8	1.8	0.1	1.2	83.7	13.2	0.9	0.3	1.9
Chemical Products	64.9	26.2	2.5	1.5	3.2	71.4	16.9	3.7	3.5	1.3
Iron and steel	75.5	16.4	0.0	7.6	0.0	29.0	47.3	0.0	22.3	0.0
Nonmetallic products	67.0	9.9	0.4	5.0	1.2	64.8	9.2	0.4	6.2	0.4
General machinery	43.4	47.6	2.1	1.7	5.1	49.0	47.8	1.7	1.1	0.4
Electric machinery	26.6	53.8	1.1	9.2	9.3	36.6	46.7	1.2	15.4	0.1
Transport machinery	55.1	39.3	1.3	2.5	1.1	52.9	43.8	1.8	1.1	0.4
Precision instruments	22.7	58.3	0.9	0.8	17.4	34.2	60.2	1.9	3.7	0.0
Coal and petroleum products	86.8	10.3	0.9	2.0	0.0	92.6	3.8	1.1	2.5	0.0
Other manufacturing	61.8	24.2	1.7	6.2	5.5	58.6	27.5	4.0	6.2	2.1

	Sales 1992									
	World					Asia				
		exports to					exports to			
	Local sales	Japan	North America	Asia	Europe	Local sales	Japan	North America	Asia	Europe
Manufacturing total	76.7	6.3	2.7	3.6	9.9	66.1	15.8	3.7	11.2	2.0
Food	63.7	20.8	2.4	2.8	3.5	46.0	26.5	3.1	4.9	2.0
Textiles	58.5	11.3	6.6	10.4	9.9	56.1	14.2	7.2	12.3	6.7
Wood and pulp	47.7	32.5	11.1	2.4	4.1	50.2	47.2	2.4	0.3	0.0
Chemical products	64.5	4.9	3.0	10.1	15.4	64.7	4.9	0.4	28.6	0.2
Iron and steel	95.8	0.8	2.1	0.8	0.1	85.5	2.1	3.0	8.6	0.9
Nonmetallic products	67.7	18.1	0.4	12.2	1.4	63.3	21.4	0.1	14.9	0.0
General machinery	72.3	4.2	5.1	2.1	15.3	53.0	23.6	2.1	11.3	9.8
Electric machinery	60.6	9.3	3.3	6.1	19.9	45.7	27.2	5.3	19.0	2.2
Transport machinery	94.1	1.4	1.8	0.4	1.9	92.6	1.7	3.9	1.0	0.5
Precision instruments	71.4	21.3	2.7	0.7	3.7	36.9	51.8	5.2	1.9	3.8
Coal and petroleum products	58.1	0.8	38.3	2.0	0.8	55.9	0.0	43.9	0.2	0.0
Other manufacturing	80.7	4.4	1.7	1.8	1.1	78.6	9.4	2.6	5.6	2.8

(*continued*)

Table 7 (*continued*)

	Procurements 1998									
	World					Asia				
		imports from					imports from			
	Local procure-ments	Japan	North America	Asia	Europe	Local procure-ments	Japan	North America	Asia	Europe
Manufacturing total	46.9	36.6	3.2	8.1	4.6	43.9	34.8	1.6	18.2	0.6
Food	81.7	4.1	1.8	10.0	0.3	78.8	6.6	0.4	8.0	0.5
Textiles	56.2	23.2	3.1	9.5	2.4	52.9	26.1	2.9	11.2	0.6
Wood and pulp	94.5	2.2	0.3	1.3	1.7	76.8	12.4	1.0	7.7	2.1
Chemical products	64.0	20.7	4.0	5.2	4.7	54.4	18.6	7.1	14.8	2.2
Iron and steel	63.2	29.9	1.9	3.7	0.2	19.2	70.0	0.0	10.4	0.2
Nonmetallic products	71.7	13.2	1.5	6.2	6.1	44.1	31.7	0.3	19.0	1.1
General machinery	38.6	44.1	2.6	2.8	11.3	57.7	32.2	0.8	8.8	0.4
Electric machinery	36.8	44.1	1.0	15.4	2.4	35.7	37.0	0.4	26.4	0.2
Transport machinery	50.2	35.8	6.1	2.0	5.6	53.7	37.0	2.5	6.0	0.7
Precision instruments	39.0	45.2	2.1	12.3	1.2	40.2	41.2	2.6	14.5	1.5
Coal and petroleum products	21.0	15.2	3.2	27.3	33.2	21.7	18.0	10.3	45.5	3.9
Other manufacturing	49.6	32.1	2.2	8.2	7.4	52.8	29.7	1.8	14.2	0.9

	Sales 1998									
	World					Asia				
		exports to					exports to			
	Local sales	Japan	North America	Asia	Europe	Local sales	Japan	North America	Asia	Europe
Manufacturing total	70.0	9.6	4.7	5.7	8.8	49.8	25.1	4.5	16.6	2.6
Food	76.7	11.6	1.4	2.5	7.0	69.1	16.1	3.3	6.4	3.5
Textiles	57.6	16.6	8.4	8.7	7.2	47.7	22.2	10.9	12.0	5.5
Wood and pulp	37.7	34.4	12.1	7.9	6.4	58.9	26.7	2.1	10.9	0.0
Chemical products	76.1	4.5	2.2	6.3	9.2	72.4	7.6	0.6	16.6	1.7
Iron and steel	91.7	1.3	2.2	2.2	0.7	85.3	3.0	2.7	6.6	0.1
Nonmetallic products	63.0	14.6	2.8	9.5	9.9	55.9	15.6	0.9	26.3	1.0
General machinery	65.4	10.8	4.6	4.4	12.3	32.4	40.6	5.5	14.8	4.6
Electric machinery	57.9	15.4	4.0	11.5	10.0	32.3	33.0	5.3	24.8	2.9
Transport machinery	81.2	2.6	6.6	0.5	8.6	81.1	11.0	3.5	2.2	1.5
Precision instruments	46.5	27.6	5.9	13.1	5.2	27.2	46.0	1.5	23.0	2.0
Coal and petroleum products	18.9	63.3	0.1	1.6	15.0	21.2	65.7	0.0	2.9	10.2
Other manufacturing	81.3	7.2	3.0	2.3	5.0	59.8	24.9	4.1	8.9	1.8

Source: Ministry of International Trade and Industry, *Kaigai jigyo katsudo kihon chosa* [Comprehensive Survey of Overseas Activities of Japanese Firms], nos. 5 and 7, 1994 and 2001.

Table 8

Intrafirm Transactions in Procurements and Sales of Foreign Affiliates of Japanese Firms: 1992 and 1998 (percentage share in each procurement or sale)

	Procurements 1992									
		World					Asia			
		imports from					imports from			
	Local procure-ments	Japan	North America	Asia	Europe	Local procure-ments	Japan	North America	Asia	Europe
Manufacturing total	9.0	84.3	52.6	58.9	68.6	4.2	78.0	47.7	50.2	35.8
Food	5.4	93.1	33.8	54.3	0.9	0.2	75.8	14.3	48.8	0.0
Textiles	15.1	37.1	3.2	31.5	2.7	19.5	34.2	3.3	31.5	14.3
Wood and pulp	6.3	30.1	0.0	0.0	0.0	0.1	79.4	0.0	0.0	0.0
Chemical products	13.5	81.7	31.8	4.6	50.8	18.0	57.5	7.7	4.3	17.1
Iron and steel	0.7	2.0	0.0	0.0	0.0	5.2	1.1	0.0	0.0	0.0
Nonmetallic products	8.4	67.6	7.3	1.0	78.3	0.0	26.0	0.0	0.0	0.0
General machinery	28.7	90.8	32.5	78.7	60.1	4.5	93.9	80.3	84.8	23.9
Electric machinery	16.6	76.0	69.0	62.6	77.3	2.0	84.6	86.6	59.8	98.1
Transport machinery	3.3	98.6	72.2	90.3	19.0	0.6	81.7	76.2	34.6	86.2
Precision instruments	9.9	74.9	93.4	15.2	98.4	17.5	85.6	0.0	100.0	0.0
Coal and petroleum products	0.0	100.0	100.0	10.0	0.0	0.0	100.0	100.0	10.0	0.0
Other manufacturing	4.3	72.2	5.9	22.5	46.4	7.3	64.1	0.7	61.8	11.8

Procurements 1998

Manufacturing total	22.2	92.3	39.8	59.9	40.7	11.3	83.2	72.5	58.5	45.3
Food	13.6	93.3	56.1	44.0	9.1	3.9	94.4	48.8	71.7	15.1
Textiles	13.1	87.0	49.6	52.7	48.9	22.1	85.1	40.5	54.8	14.2
Wood and pulp	37.4	60.5	20.6	0.0	0.0	3.5	53.8	20.6	0.0	0.0
Chemical products	16.9	86.5	71.7	77.9	47.7	17.6	76.8	85.4	79.5	82.6
Iron and steel	13.6	85.6	0.0	100.0	0.0	42.6	85.3	0.0	39.9	0.0
Nonmetallic products	29.3	83.2	10.4	39.5	16.0	20.3	78.9	58.1	37.9	1.4
General machinery	8.9	94.4	41.7	79.7	76.6	3.9	93.5	21.2	89.7	0.0
Electric machinery	19.2	91.2	39.1	59.6	51.4	8.6	80.8	32.8	58.8	10.8
Transport machinery	29.8	95.1	35.3	82.0	16.3	13.4	86.7	100.0	66.2	62.5
Precision instruments	28.9	95.7	13.3	34.3	94.4	39.1	95.5	0.0	28.3	99.5
Coal and petroleum products	59.5	2.0	0.0	9.4	0.0	74.6	1.7	0.0	9.4	0.0
Other manufacturing	13.7	90.6	38.3	35.3	77.6	6.1	81.2	48.0	57.3	34.2

(*continued*)

Table 8 (*continued*)

	Sales 1992									
	World					Asia				
		exports to					exports to			
	Local sales	Japan	North America	Asia	Europe	Local sales	Japan	North America	Asia	Europe
Manufacturing total	17.4	78.3	47.5	43.9	37.8	6.3	84.2	62.4	44.4	47.6
Food	5.2	84.6	27.2	17.2	18.6	7.6	85.4	51.9	26.3	50.1
Textiles	3.1	40.1	1.1	20.9	21.2	4.3	36.1	1.1	23.0	0.9
Wood and pulp	2.3	80.9	0.0	0.0	0.0	0.0	57.9	0.0	0.0	0.0
Chemical products	9.2	50.4	52.3	9.5	30.8	2.4	49.0	11.5	3.2	17.6
Iron and steel	0.0	16.2	3.2	0.0	0.0	0.0	29.0	23.3	0.0	0.0
Nonmetallic products	7.8	82.6	73.5	53.0	18.9	0.8	82.6	0.0	55.1	0.0
General machinery	18.3	91.2	61.6	52.8	75.2	3.0	96.7	54.3	55.6	93.9
Electric machinery	17.2	86.2	49.5	52.7	34.8	8.0	90.0	82.6	53.7	58.0
Transport machinery	24.5	49.0	67.8	75.0	42.9	7.2	73.9	71.2	57.9	28.3
Precision instruments	7.5	95.1	39.2	74.1	42.2	32.4	96.5	51.1	77.9	50.8
Coal and petroleum products	0.0	100.0	0.0	90.1	100.0	0.0	0.0	0.0	0.0	0.0
Other manufacturing	5.5	62.0	28.3	51.4	28.6	6.3	67.0	25.3	49.8	18.4

	Sales 1998									
Manufacturing total	20.4	94.6	50.6	56.7	40.5	16.1	95.7	66.4	58.9	66.1
Food	10.7	78.6	18.6	55.8	9.5	26.1	85.7	15.9	73.3	18.0
Textiles	5.8	84.2	62.0	52.6	48.0	11.2	83.4	60.0	52.2	66.5
Wood and pulp	4.2	87.3	35.3	20.5	0.0	16.3	76.6	30.4	42.4	0.0
Chemical products	16.5	92.1	37.5	27.2	23.1	12.0	91.3	42.5	25.4	38.3
Iron and steel	1.5	74.3	42.2	29.1	90.9	3.1	95.4	74.0	34.6	33.5
Nonmetallic products	35.4	90.4	13.9	32.3	5.7	14.8	99.3	98.9	32.8	94.4
General machinery	11.3	98.1	90.8	81.5	69.8	10.8	98.5	95.0	94.4	99.2
Electric machinery	16.0	96.5	49.1	64.1	53.1	20.5	96.3	55.3	66.4	66.1
Transport machinery	33.8	94.0	45.6	59.9	21.4	11.5	96.9	98.6	56.7	94.7
Precision instruments	24.6	98.4	50.0	21.8	54.0	58.3	98.4	24.2	19.1	49.6
Coal and petroleum products	6.2	90.8	0.0	0.0	0.0	6.9	91.3	0.0	0.0	0.0
Other manufacturing	10.7	92.1	41.1	31.3	46.1	8.1	94.3	62.7	32.7	29.4

Source: Ministry of International Trade and Industry, *Kaigai jigyo katsudo kihon chosa* [Comprehensive Survey of Overseas Activities of Japanese Firms], nos. 5 and 7, 1994 and 2001.

regional arrangements in the form of FTAs. This section examines such recent developments.

East Asia has been witnessing the emergence and proliferation of FTAs in recent years. One notable characteristic of FTAs in East Asia is their comprehensiveness in coverage. As such, some of the FTAs established in East Asia are called an Economic Partnership Agreement (e.g., Japan-Singapore EPA, JSEPA) or Comprehensive Economic Partnership Agreement (e.g., China–Hong Kong CEPA). These new types of FTAs typically include facilitation of foreign trade, liberalization and facilitation of foreign direct investment (FDI), and economic and technical cooperation, in addition to trade liberalization, which is included in traditional FTAs. It may be worth noting that the basic philosophy of these new types of FTAs is similar to that of the Asia-Pacific Economic Cooperation forum, whose three pillars are (1) liberalization, (2) facilitation of foreign trade and foreign investment, and (3) economic and technical cooperation.

East Asia was not active in the formation of regional trade agreements, which include FTAs and customs union, until recently (Table 9).[6] Indeed, the ASEAN Free Trade Area was the only major FTA until Japan and Singapore enacted the JSEPA in 2002. AFTA was established in 1992 with six ASEAN member countries, Indonesia, Malaysia, the Philippines, Singapore, Thailand, and Brunei. New ASEAN members—Cambodia, Laos, Myanmar, and Vietnam—joined AFTA in the latter half of the 1990s. Currently AFTA has ten member countries. Besides AFTA, ASEAN as a group as well as its members individually have become active in FTA discussions with other countries in recent years. One of the FTAs involving ASEAN that has received the most attention recently is that with China. ASEAN and China started FTA negotiations in January 2003 with the target of concluding the negotiations by June 2004. ASEAN is also discussing the possibility of FTAs with Japan and Korea. Some ASEAN members have become active in establishing bilateral FTAs. Singapore enacted or signed several FTAs with countries such as New Zealand, Japan, Australia, the United States, and the European Fair Trade Association (EFTA), and began negotiations with countries including India. Thailand has also become active in establishing FTAs, as it is currently under negotiations with the United States and Japan. The Philippines and Malaysia began negotiations with Japan in 2004.

Compared to ASEAN countries in Southeast Asia, the economies in Northeast Asia including China, Japan, Korea, and Taiwan have not been active in FTAs until recently. Despite growing interest in FTAs by Northeast Asian countries, there is only one FTA (JSEPA) that has been enacted so far. Japan is moving to set up an FTA with Mexico, and is currently negotiating with South Korea, Thailand, Malaysia, and the Philippines. Furthermore, it has been studying possible FTAs with ASEAN, Indonesia, and Vietnam. South Korea started having an interest in FTAs before Japan. In 1998 South Korea disclosed a plan to start FTA negotiations with Chile, and it also set up a joint-study group at the private level on an FTA with Japan. South Korea started negotiations with Chile in 1999, and signed the agree-

Table 9

Selected Free Trade Agreements in East Asia (as of June 2004)

Action	Negotiation	Study
Bangkok Treaty (1976)	China-ASEAN	Japan-ASEAN
AFTA (1992)	Hong Kong-New Zealand	Korea-ASEAN
Singapore-New Zealand (2001)	Japan-Mexico	Korea-Australia
Japan-Singapore (2002)	Japan-Korea	Korea-New Zealand
Singapore-EFTA (2003)	Japan-Thailand	Singapore-Chile
Singapore-Australia (2003)	Japan-Philippines	Singapore-Taiwan
China-Hong Kong (2004)	Japan-Malaysia	Thailand-Peru
Singapore-United States (2004)	Korea-Singapore	ASEAN-India
Korea-Chile (2004)	Singapore-Canada	ASEAN-United States
	Singapore-Mexico	
	Singapore-India	
	Taiwan-Panama (concluded)	
	Thailand-Bahrain (concluded)	
	Thailand-India (concluded)	
	Thailand-United States	
	Thailand-Australia	

Source: Author's compilation from national sources.

ment in October 2002 after difficult negotiations on the liberalization of agricultural imports. After reaching an agreement, it took the Korean National Assembly more than a year to ratify it, because of strong opposition from farmers. South Korea also started studying the possibility of FTAs with Singapore, New Zealand, and ASEAN.

China's active FTA strategy has received a lot of attention. China joined the WTO in 2001, establishing access to the world market. It also started to pursue regional strategies by using FTAs. China signed a framework agreement on comprehensive economic cooperation with ASEAN in November 2002. The agreement, which was proposed strongly by China, covers not only trade liberalization but also cooperation in the areas of FDI and economic development. As noted above, China and ASEAN started negotiations on an FTA in 2003. China has offered various schemes attractive to ASEAN and particularly to its new members such as economic cooperation for the new ASEAN members and advanced trade liberalization (early harvest) in agricultural products. In addition to ASEAN, China has proposed to Japan and South Korea to establish a trilateral FTA. China enacted an FTA (CEPA) with Hong Kong.

The idea of an FTA covering East Asian countries has emerged. At the 1998 summit meeting of ASEAN + 3 (China, Japan, and South Korea), the leaders de-

cided to set up the "East Asia Vision Group" to study a long-term strategy for economic cooperation. The group has presented the leaders with recommendations including the establishment of an East Asia FTA. Despite the recommendation from the Vision Group, the East Asia FTA has not yet become a concrete agenda at the leaders meeting.

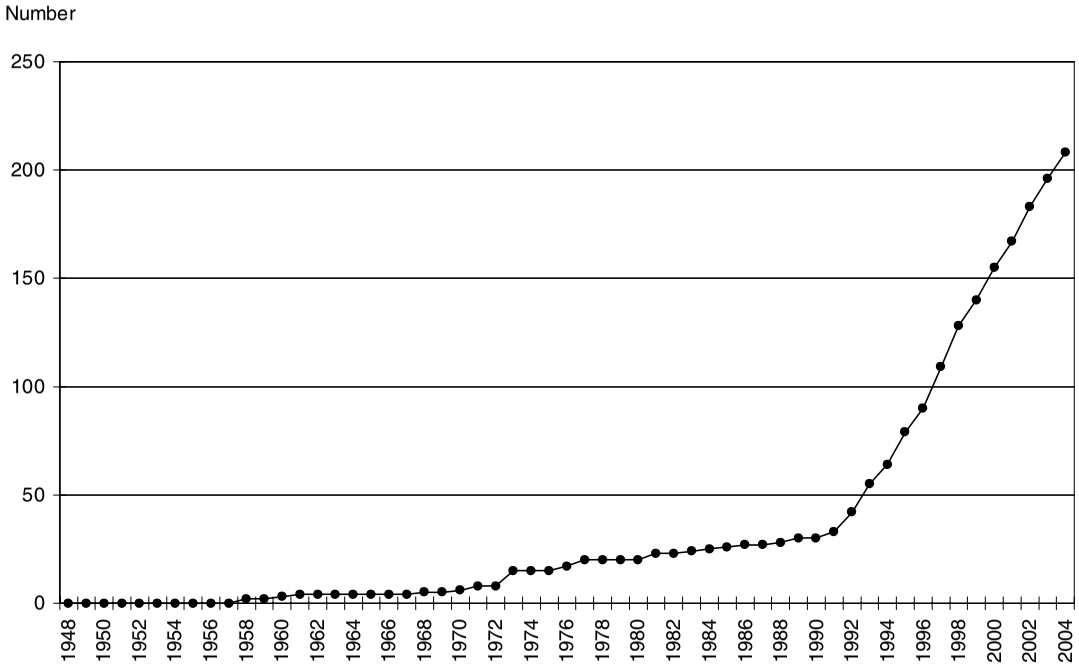
The Factors Behind the Proliferation of Free Trade Agreements in East Asia

Various factors can be identified behind the emergence of a strong interest in FTAs in East Asia. First, rapid expansion of FTAs in other parts of the world has prompted East Asian economies to have an interest in establishing FTAs to maintain and expand market access for their exports (Figure 3). Many East Asian economies have begun to realize the benefits of trade liberalization for the promotion of economic growth, as it has brought about the expected outcome. Despite the desire of many policy makers to promote further trade liberalization, it is apparent that progress under the GATT/WTO has become increasingly difficult. As the number of GATT/WTO members has increased, views on the pace and the extent of trade liberalization have diverged. Indeed, the fact that it took the GATT members eight years, twice as long as initially planned, to conclude the Uruguay Round (UR) negotiations, indicates the increasing difficulty in reaching a consensus on trade liberalization. This difficulty was a factor in the failure of the WTO members to initiate a new round of trade negotiations in Seattle in 1999. Although an agreement was reached in Doha to launch a new round, substantive negotiations have not proceeded apace, as even the modality of the negotiations has yet to be agreed on. Faced with difficulty in carrying out trade liberalization on the global scale, many countries in other parts of the world have opted to form FTAs among like-minded countries to pursue trade liberalization. As a result of a rise in FTAs in other parts of the world, East Asian countries started to feel discriminated against in many markets in the world. To overcome such a disadvantage and to secure markets for their exports, East Asian countries have become active in forming FTAs.

It should also be noted that many countries in the world, including those in East Asia, began to realize that GATT/WTO rules cannot adequately deal with newly emerging international economic activities such as FDI, services trade, and mobility of labor. To put it differently, rules concerning border measures such as tariffs, which are the main focus of the GATT/WTO, cannot provide foreign as well as domestic companies with a level-playing field. It is necessary to go deeper beyond the border measures and to set up rules covering domestic systems such as competition policy. Many countries share the view that GATT/WTO cannot provide such rules and thus have opted for using FTAs to deal with the problem.

It is important to note that East Asian countries recognize that the EU and the NAFTA are very successful in promoting economic growth, and thus East Asian countries have become interested in forming FTAs. East Asia's interest in FTAs

Figure 3. **Regional Trade Agreements (RTAs) in the World: Cumulative Number of RTAs Reported to General Agreement on Tariffs and Trade/World Trade Organization as of May 1, 2004**



has intensified as the EU and countries in North and South America expand their FTA networks.

Second, somewhat related to the point just made, East Asian economies have become interested in using FTAs as a way to promote deregulation and structural reforms in the domestic market. Many East Asian economies pursued deregulation and structural reforms in the 1990s, which contributed to the realization of rapid economic growth. Although many East Asian economies recognize the need for accelerating deregulation and structural reforms to further promote economic growth, they have found that doing so may not be easy for them as many economies are still recovering from the crisis, and as the room for further deregulation became limited as a result of earlier deregulation. Under these circumstances, external pressures such as FTAs are considered very effective in pursuing deregulation.

Third, many East Asian economies have become interested in using FTAs to promote economic and other types of cooperation in East Asia. The financial crisis in the late 1990s in East Asia increased the awareness of the need for regional cooperation such as FTAs to avoid another crisis and to promote regional economic growth because East Asian economies could not get as much assistance as they hoped from countries outside the region. Indeed, as I have noted, FTAs and future FTAs in East Asia include not only trade liberalization under the traditional approach but also trade and FDI liberalization and facilitation under the name of EPAs. It should be noted that regional cooperation in the area of finance has moved forward significantly, as many East Asian economies realized the need for such cooperation to avoid another financial crisis. Specifically, the Chiang-Mai Initiative, which consists of bilateral currency swap arrangements, was instituted by selected East Asian countries.

Fourth, it has to be emphasized that political factors have contributed to the increased interest in FTAs in East Asia. A rivalry between China and Japan for becoming a "leader" in East Asia has made them interested in using FTAs to strengthen their relationships with ASEAN and the NIEs. Indeed, in November 2002 Japan proposed an economic partnership framework to ASEAN one day after China agreed to start FTA negotiations with ASEAN. It should also be noted that ASEAN and the NIEs consider FTAs as a means to maintain and increase their influential position in East Asia. This is particularly notable for ASEAN, as it has been seeking to establish FTAs with China, Japan, and Korea separately, or three "ASEAN + 1" FTAs, to maintain and increase its influence before the establishment of an East Asia-wide FTA, which will be dominated by China and Japan. ASEAN has been successful in attracting India, the United States, the EU, and other countries outside East Asia to form FTAs, mainly to maintain its influential position in East Asia. It should be noted that India, the United States, the EU and other countries outside the region regard the relationship with ASEAN countries important to maintain their presence in East Asia, in light of the emergence of regionalization in East Asia.

South Korea, one of the NIEs, sandwiched between China and Japan not only

geographically but also economically and politically, is keen on maintaining its position as a balancer between the two countries. Indeed, South Korea has been an active advocate of an FTA involving China, Japan, and itself. It should also be noted that South Korea is interested in establishing a cooperation mechanism in Northeast Asia with China and Japan, which would potentially facilitate the unification of South and North Korea.

Taiwan stepped up its efforts to establish FTAs with various countries after becoming a WTO member in 2002. Taiwan is interested in using FTAs not only to expand economic relationships but also to establish political and diplomatic relationships. Despite its desire to form FTAs with many countries, its diplomatic status in the international community has made it difficult to achieve its objectives so far.

Finally, we should point out the reasons for the emergence of EPAs rather than traditional FTAs in East Asia. One notable characteristic of East Asia is diversity in the level of economic development in the region, ranging from high-income countries such as Japan and Singapore to low-income countries such as the new ASEAN members. In order for FTAs involving such diverse countries to become a success, traditional trade liberalization under FTAs should be supplemented by cooperation programs such as trade and FDI facilitation and economic assistance under EPAs, as such programs reduce the adjustment costs of trade liberalization.

Benefits of Free Trade Agreements in East Asia

The countries interested in FTAs expect various benefits from the formation of FTAs, ranging from economic to noneconomic benefits. This section first discusses benefits of FTAs in terms of trade liberalization and then turns to additional benefits from deeper integration through implementing additional features such as FDI liberalization and economic cooperation under new types of FTAs.

Let us first examine the possible impacts of an FTA on member countries through foreign trade. The trade impacts of an FTA can be classified into two types. One is the static effect and the other is the dynamic effect. As to the static effect, trade creation, trade diversion, and terms of trade effects are important. Removal of trade barriers among the members would promote trade among them (trade creation effect) sometimes at the expense of imports from nonmembers (trade diversion effect). If an FTA leads to a reduction in imports from nonmembers, FTA members are likely to experience an improvement in their terms of trade vis-à-vis nonmembers (terms of trade effect). The trade creation effect and the terms of trade effect lead to an increase in the economic welfare of members, while the trade diversion effect is likely to reduce the economic welfare of members because imports from the most efficient suppliers in nonmember economies are replaced by imports from less-efficient producers in member economies. It is important to note that for nonmembers, the trade diversion effect and terms of trade effect lead to harmful impacts on their economic welfare.

The preceding discussions on the impact of FTA on foreign trade are given from the point of view of an economy as whole. From the point of view of the business sector, which has strong influence on the policy-making process, two different types of effects are expected. For competitive business, FTA will bring benefits as it enables them to increase their access to members' markets. However, for the noncompetitive business sector, FTA will have a harmful effect as it introduces competitive pressure.

In addition to these static effects, dynamic effects can be expected. They include mainly a scale effect and a competition-enhancing effect, both of which would increase economic welfare for members and nonmembers. The scale effect may arise from the formation of a larger market as a result of an FTA. Faced with a larger market, companies may be able to exploit the benefits of economies of scale, which would benefit consumers in both member and nonmember countries. A procompetition effect may be expected as competition in a member country's market is likely to intensify as a result of trade liberalization. The procompetition effect may give rise to new products and new technology, which in turn would benefit consumers in both member and nonmember countries. One may consider the procompetition effect as the effect that would promote domestic policy reform. Many countries can use external pressure to promote domestic reform. In the past, GATT commitments as well as threats from the United States and other countries represented effective external pressures. However, at present, these pressures are weak. Under such circumstances, FTAs could be effective external pressure.

In addition to the effects on foreign trade, FTAs are likely to impact FDI. Specifically, one would expect both an FDI creation effect and an FDI diversion effect. An FDI creation effect indicates increased FDI among member countries, while an FDI diversion effect refers to a shift of FDI from nonmember to member countries. Both effects have positive impacts on member countries in that FDI would flow into member countries, as trade liberalization via FTA would increase the attractiveness of members' markets. While the FDI creation effect does not have any immediate negative impacts on nonmember countries, the FDI diversion effect has negative impacts as FDI is diverted from nonmember to member countries. So far I have discussed only the indirect impact of free trade agreements on FDI. If an agreement includes FDI liberalization, as is seen under the new type of FTAs, then naturally it would directly yield the kinds of impacts on FDI discussed above.

The preceding discussion indicates possible benefits for member countries and possible negative impacts on nonmember countries. Having noted this, I would like to emphasize that the above statement is correct only if the static effects of FTAs are considered. If both static and dynamic effects are taken into account, the FTA is likely to bring benefits not only to members but also to nonmembers.

Let us examine the possible impacts of an eleven-member East Asia FTA on East Asian economies. Urata and Kiyota (2003) conducted a simulation exercise

by using a general computable equilibrium (CGE) model. Table 10 presents their results for GDP and welfare, measured by equivalent variation (EV), for East Asian economies with a few selected countries. The results indicate that all FTA member economies gain benefits from an East Asia FTA in terms of GDP and EV. The positive impacts are very large for the ASEAN countries. Among the ASEAN countries, Thailand gains substantially. Indeed, Thai GDP is estimated to increase as much as 16 percent due to an East Asian FTA. A large gain for Thailand is attributable mainly to the high protection imposed on the Thai economy prior to the formation of an East Asia FTA. Vietnam and Indonesia also would gain substantially from the East Asia FTA.

In contrast to the gains accrued by FTA members, nonmember countries experience negative impacts in the form of declines in GDP and EV. These negative impacts are mainly attributable to the trade diversion effect from the East Asia FTA, through which nonmember countries' exports to East Asia are substituted by member countries' exports as a result of preferential treatment given to trade involving members. It should be noted that the negative impacts on the United States and the EU are quite small while they are somewhat substantial for Australia/New Zealand and Asian economies ("other Asia" in the table) outside the hypothetical FTA. Relatively large negative impacts for Australia/New Zealand and "other Asia" are attributable to the fact that East Asia is a very important region for their exports.

Having presented the estimates based on a CGE model, it is important to note that these estimates are likely to be smaller than actual impacts, because they do not fully incorporate dynamic effects. In addition, the impacts would be greater if deeper integration, including trade and FDI facilitation, as well as economic cooperation, is adopted.

Before ending the discussion on the benefits of FTAs and EPAs to member countries, I should emphasize the noneconomic benefits from these arrangements. Through EPAs, member countries expect to deepen mutual understanding, which would reduce political and social frictions to achieve social and political stability. Social and political stability in turn would contribute to economic growth.

Obstacles to Free Trade Agreements in East Asia

One may classify the obstacles to the formation of FTAs in East Asia into several categories including economic, historical, and political factors. Let me begin with economic factors. The formation of an FTA with countries of different levels of economic development, as is the case for East Asia, is likely to result in sizable economic benefits as the removal of trade barriers through FTA enables the members to use resources efficiently by making them specialize in the production of goods with a comparative advantage. Theoretically, this point may be used as a rationale for establishing FTAs in East Asia, particularly an East Asia-wide FTA, but in reality the formation of such an FTA would encounter strong opposition from

Table 10

Effects of East Asia Free Trade Agreements on Real GDP and Equivalent Variation

	GDP	Equivalent variation	
	Changes from, base data, %	Changes from base data, US$million	Changes divided by GDP in 1997, %
Australia/New Zealand	–0.23	–1,342	–0.29
China	1.27	5,485	0.64
Hong Kong	1.41	3,389	2.42
Japan	0.05	8,199	0.19
Korea	1.71	7,805	1.75
Taiwan	1.51	5,597	1.87
Indonesia	5.61	10,209	4.89
Malaysia	2.83	2,279	2.15
Philippines	2.02	602	0.77
Singapore	2.26	2,944	3.69
Thailand	15.90	19,790	12.54
Vietnam	8.42	1,446	6.61
Other Asian nations	–0.31	–1,803	–0.34
United States	–0.06	–7,059	–0.09
European Union	–0.01	–1,807	–0.02

Source: Shujiro Urata and Kozo Kiyota, "The Impacts of an East Asia FTA on Foreign Trade in East Asia" (paper presented at the Fourteenth Annual East Asian Seminar on Economics, Taipei, Taiwan, September 5–7, 2003).

noncompetitive sectors, which are pressured to go through painful structural adjustment. Indeed, it may be argued that the formation of FTAs involving countries with a similar level of development is easier because they are engaged more actively in a horizontal type of intraindustry trade, and under such circumstances weaker opposition may be expected.

This point may be supported by the results of simulation exercises by Urata and Kiyota (2003). According to their estimates, shown in Table 11, the level of production in noncompetitive and "sensitive" sectors is projected to decline as a result of the FTA. For example, an East Asia FTA would result in a reduction in the output of agriculture in Japan and South Korea, and transportation machinery in China, Taiwan, Malaysia, Indonesia, and several other countries. Although agriculture is a declining industry for Japan and Korea, and therefore it may be better to reduce the size of agricultural production from the point of view of resource

allocation, a strong opposing coalition of farmers, politicians, and bureaucrats makes it difficult for these governments to pursue an East Asia FTA. Many developing East Asian countries, which are interested in developing the presently noncompetitive automobile sector, would not like to see a decline in that sector.

In addition to these economic factors, several political factors are hindrances to FTAs in East Asia. One is a security issue. East Asia has countries with different political systems. While most countries have a democratic political system, China and Myanmar have authoritarian regimes. While many countries have security alliances with the United States, China is still considered a possible threat to these alliances. These issues may be dissolved as economic development is achieved and as international exchange among these countries is enhanced, resulting in sharing common views toward political systems and regional security.

Another political problem impeding the establishment of an East Asia FTA is the absence of strong political leadership pushing its formation. The experience in Europe shows the crucial role that strong political leadership played in the formation of the European Economic Community in 1957 and the subsequent deepening of integration. It is reasonable to assume that China and Japan should jointly take leadership in East Asia, but so far the differences in their views on various issues such as political systems and regional security, as noted above, have been obstacles.

Finally, different views on history involving the Northeast Asian countries, namely, China, Japan, and Korea, are obstacles to an East Asia FTA. Closer economic and social relations will contribute to reducing the gaps in their views on historic and other issues.

Conclusions: Suggestions to Overcome the Obstacles Toward the Establishment of an East Asian Free Trade Agreement

East Asian countries have been active in forming FTAs with other East Asian countries. Many East Asian countries consider the formation of an East Asia FTA as a tentative goal and deeper economic integration such as the EU as a long-term goal. The motive behind these policies is potential economic and noneconomic benefits from these arrangements. However, various obstacles do exist as was discussed in the previous section. In this section some possible ways to overcome these obstacles will be presented.

One of the obstacles has to do with the structural adjustment that is required in the formation of FTAs. More specifically, noncompetitive sectors have to face increased competitive pressure from FTA partners. One possible way to deal with the needed structural adjustment is to implement scheduled trade liberalization in sensitive sectors as a part of an FTA. GATT/WTO allows the members of an FTA to take ten years to complete the FTA. East Asian countries should use this breathing space to facilitate structural adjustment through appropriate adjustment policies. For example, affected workers should be given financial, technical, and other types of assistance so that they can improve their quality of human resources in order to be able to

Table 11

Effects of East Asia Free Trade Agreements on Production by Sectors (% change)

Sector	China	Japan	Korea	Hong Kong	Singapore	Taiwan	Indonesia	Malaysia	Philippines	Thailand	Vietnam
Agriculture	4.5	–3.9	–11.9	0.3	4.7	1.6	1.3	0.4	2.0	5.2	0.4
Forestry	–0.2	–1.8	–3.6	1.7	–7.4	3.2	8.6	2.1	2.1	16.3	11.4
Fishing	0.9	–2.8	7.6	–2.6	7.8	–1.0	3.9	2.5	1.3	12.6	6.1
Mining	–0.2	–0.7	–2.1	3.8	4.1	–0.9	1.4	1.7	4.1	19.2	–8.3
Food products and beverages	1.6	–3.4	30.1	19.3	36.7	6.9	5.3	15.3	–1.3	23.5	8.9
Textiles	–0.2	–2.3	17.4	3.2	1.9	17.1	3.7	7.7	13.9	8.4	174.8
Pulp, paper, and paper products	–1.1	–0.4	1.9	3.3	3.7	1.6	8.9	4.3	0.8	16.1	16.9
Chemicals	–1.6	1.1	3.4	5.5	11.1	7.1	1.4	4.4	2.0	10.6	–2.0
Iron, steel, and metal products	–1.5	2.2	–1.4	4.7	7.7	0.0	2.9	1.4	6.7	20.1	–18.3
Transportation machinery	–16.2	5.2	1.0	–7.9	–14.3	–6.9	–47.8	–24.0	29.3	–11.0	–55.1
Electronic equipment	6.9	–0.7	–1.9	0.9	1.9	–2.5	17.4	5.7	8.9	29.2	–0.9
General machinery	–1.6	2.2	–4.8	7.4	5.5	1.7	22.8	7.4	12.7	26.8	–3.7
Other manufacturing	1.6	–0.5	0.9	8.1	5.0	2.5	7.3	1.2	5.8	18.1	12.9

Source: Shujiro Urata and Kozo Kiyota, "The Impacts of an East Asia FTA on Foreign Trade in East Asia" (paper presented at the Fourteenth Annual East Asian Seminar on Economics, Taipei, Taiwan, September 5–7, 2003).

Notes: The figures indicate changes from the base data.

obtain more productive jobs. If such a program is successful, trade liberalization through FTAs can be pursued smoothly to result in benefits for all the countries. Indeed, the needed assistance should be provided by members as a part of economic assistance programs under new types of FTAs. For example, in the case of educating and training workers for upgrading their skills, more developed members such as Japan, Korea, Taiwan, and Singapore can provide useful assistance to other countries. In the case of developing competitive small and medium-sized enterprises, which would ease the transition process, Japan can provide assistance to other countries. Furthermore, it is very important to analyze economic and other issues jointly to learn important policy lessons, which may be applicable to the participating countries. The need for establishing new types of FTAs should be stressed.

As repeatedly noted, East Asian countries need to deepen mutual understanding at all levels, from top leaders to young people, to increase awareness of the importance of integrated regional markets and regional political and social stability. Leaders' meetings should be held at least annually to increase mutual understanding. Frequent television conferences can be used to supplement face-to-face meetings. Policy makers should establish close communication links. Bureaucrats, business people, academics, students, and others should also increase their exchange. In order to facilitate such exchange, the establishment of frameworks such as student exchange programs is effective. Such programs can be set up independently but they will be more effective if they are coordinated under FTAs. Having discussed the need for deepening mutual understanding for the implementation of FTAs in East Asia, it should be emphasized that strong political leadership is a crucial factor that would help realize new and drastic policies such as FTAs.

Finally, East Asian countries should not regard completing integration of a regional market in East Asia under an FTA as a goal, but should regard it as a step toward achieving global free trade under the WTO. This is very important not only because an East Asian FTA has negative impacts on nonmembers such as the United States, the EU, and other countries, but also because East Asian countries benefit most from global free trade. To avoid having an East Asia FTA become a stumbling block for global free trade, East Asian countries should make every effort to promote multilateral trade negotiations under the WTO and establish FTAs with countries outside East Asia.

Notes

1. An analysis of trade and FDI draws on Kawai and Urata (2004).
2. Urata (2001) conducted similar analyses for the earlier period.
3. The figures are computed as follows. International input-output tables with twenty-four sector disaggregation and ten economies are used in the analysis. To derive the amount of output induced by an increase in final demand, the Leontief inverse matrix is multiplied by a final demand vector consisting of unity for twenty-four sectors. The resulting output is divided by twenty-four to obtain the level of output induced by one unit increase in final demand.

4. See Borrus (1999) for the case of U.S. electronics firms in East Asia.

5. Gereffi (1999) presents an interesting analysis of an apparel commodity chain developed by firms from the NIEs.

6. In the GATT/WTO, regional trade agreements, which violate one of its basic principles of nondiscrimination, are permitted under GATT Article XXIV with several conditions, which include liberalizing substantially all the trade of the members, not increasing trade barriers on nonmembers, and completing the RTA process within ten years. For developing countries, more lenient conditions are applied under the enabling clause. An FTA is considered to be a shallow form of regional integration, because it only removes tariff and nontariff barriers among the members, while a customs union is a deeper integration, as it adopts common external tariffs on nonmembers, in addition to the removal of tariff and nontariff barriers on trade among the members.

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